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# Acute subdural haematoma in the elderly- to operate or not to operate? A systematic review and meta-analysis of outcomes following surgery



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# Acute subdural haematoma in the elderly- to operate or not to operate? A systematic review and meta-analysis of outcomes following surgery

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## Abstract

**Objectives**: Acute subdural haematoma (ASDH) is a devastating pathology commonly found on CT brain scans of traumatic brain injury (TBI) patients. The role of surgical intervention in the elderly has been increasingly questioned due to its associated morbidity and mortality. Therefore, a systematic review and meta-analysis of the literature to quantify the mortality and functional outcomes associated with surgical management of ASDH in the elderly was performed. **Design**: A multi-database (PubMed, Embase, Cochrane Reviews) literature search between January 1990 and May 2020 was performed. All studies reporting clinical outcomes post-surgical evacuation of ASDH (craniotomy or decompressive craniectomy) in patients aged 60 years or older were included. Meta-analysis of proportions was performed to quantify mortality and unfavourable outcome (GOS 1 – 3) rates. **Results**: 2572 articles were screened, yielding 21 studies for final inclusion and 15 for meta-analysis. Pooled estimates of mortality were 39.83% (32.73; 47.14%, 95% CI- 10 studies, 739 patients) at discharge and 49.30% (42.01; 56.61%, 95% CI- 10 studies, 55 patients) at LTFU. Average duration of follow up was 7.1 months (range 2 – 12 months). Pooled estimate of percentage of poor outcomes was 81.18% (75.61; 86.21%, 95% CI) at discharge, and 79.25% (72.42; 85.37%, 95% CI) at LTFU. Average duration of follow up was 6.4 months (range 2 – 12 months). Potential risk factors for poor outcome included age, baseline functional status, pre-operative neurological status, and imaging parameters. **Conclusions**: Outcomes following surgical evacuation of ASDH in patients aged 60 years and above are poor. This constitutes the best level of evidence in the current literature that surgical intervention for ASDH in the elderly carries significant risks, which must be weighed against benefits. This provides robust evidence that will help healthcare professional decision-making and counselling of families on a contentious issue.

**Study registration**: The study protocol is registered on PROSPERO (CRD42020189508).



**Keywords**

Traumatic brain injury, elderly, acute subdural haematoma, craniotomy


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**Strengths and limitations of this study**

- Critical appraisal of studies reporting both mortality and functional outcomes following surgical evacuation of ASDH in the elderly
- Examination of risk factors for poor outcome in included studies
- Heterogeneity arising from potential differences in indications for surgical intervention
- Potential reporting bias and heterogeneity in meta-analysis from inclusion of smaller retrospective studies

## Introduction

Traumatic brain injury (TBI) is amongst the leading causes of mortality and morbidity worldwide[1,2]. With improvements in road safety measures, we are currently experiencing an increasing contribution of traumatic injuries in the elderly population towards the incidence of major intracranial injuries[3,4,5]. Statistical data from the Centres for Disease Control and Prevention (CDC) demonstrated a 17% increase in accidental falls-related TBI between 2008 and 2017 in the USA [6]. Importantly, mortality rates secondary to falls-related TBI were disproportionately higher in patients aged 75 years and above, at 54.08 per 100,000 patients. With estimates of approximately 1.5 billion individuals aged above 65 years by 2050, there is an urgent need for effective and pragmatic management of elderly patients with TBI[7]. Acute subdural haematoma (ASDH) is a potentially life-threatening form of TBI encountered by emergency services, which is characterised by extra-axial collection of acute blood between the dura and underlying brain parenchyma. The elderly are particularly vulnerable due to the risks of falls-related TBI, age-related atrophic changes in the brain, and the use of anticoagulants[8]. Hence, rationalisation of ASDH management in this group of patients is becoming increasingly pertinent[9,10]. This was highlighted by a recent single centre retrospective study in the UK, which demonstrated that the odds of inpatient death were approximately 15 times higher than its chronic counterpart in patients aged 75 years and above [11].

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In the acute setting, ASDH management consists of either urgent surgical intervention or conservative management. The former typically consists of a craniotomy and evacuation of ASDH, though a decompressive craniectomy (DC) may be required in rare instances where the brain is severely swollen intra-operatively. However, the decision for surgical intervention for ASDH in elderly patients remains a point of contention due to its associated mortality[12]. Surgical intervention is typically guided by imaging parameters including an ASDH > 10 mm thickness or midline shift (MLS) > 5 mm on computed tomography (CT) imaging, neurological deterioration, and/or intracranial pressure (ICP) monitoring[13]. Whilst these guidelines are applicable to the general management of adult TBI, their applicability to the elderly population is less clear. Quantifying surgical morbidity and mortality is vital for tailoring decisions to this age group and managing the expectations of relatives who may not appreciate the associated risks. Therefore, in this study we perform a systematic review and meta-analysis of the literature to quantify the mortality associated with surgical management of ASDH in the elderly, and identify potential risk factors for poor outcome in this group. Herein, we demonstrate a pooled estimated mortality rate of approximately 50% and unfavourable outcome rate of approximately 80% following surgical intervention for ASDH in elderly patients. We hope that our findings guide healthcare professionals in both secondary and tertiary healthcare services on shared decision-making with patients' relatives and families to offer well-informed treatment plans.

## Methods

This study was performed in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) Statement[14]. The protocol for this systematic review is registered on PROSPERO (CRD42020189508).

### Literature Search

A multi-database (PubMed, Embase, and Cochrane Reviews) literature search was performed between January 1990 and May 2020 by authors OM and OE (**Figure 1**). Conflict of opinion was settled by senior author MZ. The following search terms were used in varying

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combinations: “subdural h(a)ematoma”, “subdural h(a)emorrhage”, “outcome*”, “compar*”, “morbid*”, “mortality*”, “complication*”, “reoccur*”, “recur*”, “surg*”, “operati*”, “non(-)surgical”, “burrhole”, “crani*”, “old*”, “frail*”, “geri*”, and “elder*”. Eligible articles were restricted to English language only, and the bibliographies of included studies were screened for further relevant studies.

**Study Selection**

Studies fulfilling the following criteria were included: (i) diagnosis of traumatic ASDH (confirmed on CT imaging), (ii) patient age of 60 years or above, (iii) patients that underwent surgical evacuation of ASDH, (iv) reported clinical outcome. Studies including patients younger than 60 years were only included if management and clinical outcomes were specifically reported for the target age group. Exclusion criteria included: (i) diagnosis of chronic subdural haematoma with no evidence of acute component; (ii) significant life-threatening extracranial injuries; (iii) abstracts, conference presentations, editorials, and expert opinions.

**Data Extraction**

The following variables of interest were extracted by authors SM and RS from eligible studies: number of patients, age, gender, baseline functional status, pre-admission residence, comorbidities, use of antithrombotic agents (antiplatelets or anticoagulants), mechanism of injury, pre-operative GCS/ pupillary reaction to light, imaging findings, indications for surgical intervention, timing of surgery, intervention performed (craniotomy or decompressive craniectomy), mortality (at discharge and longer term follow up), clinical outcome (at discharge and longer term follow up), and duration of follow up. Variables were summarised with respect to whether included studies examined associations with outcome. Studies reporting mortality and/or Glasgow Outcome scale (GOS) were considered for inclusion in meta-analysis (see below). When reported, the Glasgow Outcome Scale (GOS) was used to dichotomise outcomes into good (GOS 4 – 5) and poor (GOS 1 – 3) categories. Studies reporting conservative management along with surgical management were examined for potential comparison between different treatment groups (see below). Studies reporting minimally invasive techniques for ASDH evacuation were included for narrative synthesis. All included studies were assessed for quality using the STROBE

(Strengthening of Reporting of Observational Studies in Epidemiology) criteria[15] by two authors independently (IM and JM) (**Table 1**).

**Statistical Analysis**

Statistical analysis was performed using R Version 3.6.0, and the *metafor* and *meta* packages[16]. The percentage of patients under the following categories were pooled for meta-analysis: (i) deaths at discharge, (ii) deaths at longer term follow up (LTFU), (iii) poor outcome at discharge, (iv) poor outcome at LTFU. Poor outcome was defined as patients with GOS 1 - 3. Studies using other scoring systems were not included for meta-analysis to avoid introducing bias from definition of outcome. LTFU was defined as any time point beyond the time of discharge (specific duration of follow up was recorded). All studies reporting mortality and/or GOS (specific score or dichotomised) in patients aged 60 years or above that underwent surgical evacuation of ASDH were included for meta-analysis. Surgical evacuation consisted of either craniotomy or decompressive craniectomy.

Heterogeneity was quantified with the following tests: DerSimonian and Laird estimator[17] to estimate between-study variance ($\tau^2$), Cochran's Q test, and the $I^2$ statistic. Given that $I^2$ > 50% in all analyses and there was potential heterogeneity between studies with respect to specific age groups, a random effects model with double-arcsine transformation[16,18] was used. Pooled estimates were reported with 95% confidence intervals (95% CI). Studies published prior to January 2000 were excluded from meta-analysis to avoid distortion of summary effect sizes due to differences in healthcare technology as a function of time. Subgroup analysis was performed using the year of the study as a moderator (pre- and post-January 2000) with a mixed effects model[16]. Differences between subgroup summary estimates were significantly different. Therefore, older studies were excluded on both theoretical and statistical premises. Conservative and surgical groups were not compared due to fundamental differences in treatment indications and baseline cohort characteristics, which were acknowledged by relevant studies (**Supplementary Table 1**).

## Results

A total of 2572 articles were screened, yielding 21 studies for final inclusion [19-39](**Table 1**). A set of fifteen studies were suitable for meta-analysis, with varying subsets used for each separate analysis depending on available outcome data. All studies were single centre retrospective studies, except for a recent retrospective multicentre study performed in Italy involving 213 patients from 5 centers[37].

### Mortality

Mortality was generally defined as death by a specific time point, except for one study that also included death or discharge to a hospice[33]. Mortality at discharge (381/820 patients) and LTFU (322/622 patients) was reported by 13 studies, of which 10 studies were included for meta-analysis (**Figure 2**). Duration of follow up was provided by all studies that reported mortality at LTFU, with an average of 7.4 months (range 2 - 18 months). The pooled estimate of percentage of deaths was 39.83% (32.73; 47.14%, 95% CI) at discharge, and 49.30% (42.01; 56.61%, 95% CI) at LTFU. Average duration of follow up in studies included for meta-analysis was 7.1 months (range 2 - 12 months) (**Table 2**).

### Functional Outcomes

Poor outcome was defined as patients with GOS 1 - 3. GOS on discharge (poor outcome in 437/532 patients) was reported by 9 studies, of which 6 were included for meta-analysis. GOS at LTFU (poor outcome in 459/578 patients) was reported by 11 studies, of which 8 were included for meta-analysis (**Figure 2**). Duration of follow up was provided by all studies that reported GOS at LTFU, with an average of 6.8 months (range 2 - 18 months). Pooled estimate of percentage of poor outcomes was 81.18% (75.61; 86.21%, 95% CI) at discharge, and 79.25% (72.42; 85.37%, 95% CI) at LTFU. Average duration of follow up in studies included for meta-analysis was 6.4 months (range 2 - 12 months) (**Table 2**). One study[36] reported modified Rankin Scale (mRS) scores at discharge and LTFU. Poor outcome was defined as mRS 5 - 6, and reported in 28% (8/28 patients) at discharge and 57% (16/28 patients) at LTFU.

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## Risk factors for poor outcome

*Demographics*

Inclusion age was reported by all studies, and analysed with respect to outcome in older age groups in 9 studies (**Table 3**). Of these, 2 studies reported a significant effect of age on outcome[30,33], whilst the remainder reported no significant associations. With respect to GOS at 30 days, age was negatively associated with GOS on multivariate (RC -0.054, p = 0.013; other factors included volume of ASDH, GCS, MLS, and antiplatelet use) but not univariate linear regression (RC - 0.043, p = 0.095) analyses[30]. Age was negatively associated with GOS at 6 months on both univariate (RC -0.063, p = 0.031) and multivariate (RC -0.077, p = 0.002) linear regression analyses[30]. Another study[33] demonstrated a significant association between patient age group (70-79 years, 80 years and above) and mortality on univariate analysis (p = 0.05; Chi-squared test) and age (70-79 years vs 80 years and above) as a predictor of mortality on multivariate logistic regression (OR 2.83; 95% CI 1.18, 6.83; p = 0.02; other covariates included GCS and surgical approach). Gender was reported by 15 studies, and analysed with respect to outcome in 4 studies (**Table 3**). A significant association with outcome was not demonstrated in any of these studies[29,35,37,38].

*Baseline functional status*

Baseline functional status was reported in 11 studies, of which 5 studies examined association with outcomes (**Table 3**). One study[38] included patients aged 80 years and above demonstrated a significant association between outcome (favourable, unfavourable) and number of co-morbidities (less than or equal to 1, more than 1) (p = 0.05; Fisher's exact test). This was not statistically significant on analysis with a greater number of co-morbidities (less than 5, more than or equal to 5). In contrast, other studies reported no significant association between number of co-morbidities and dichotomised GOS[31,37] or mortality[33]. Two studies demonstrated no statistically significant differences between survivors and non-survivors with respect to pre-morbid independence[29,33] or pre-admission residence[29].

*Use of anti-thrombotic agents*

Use of anti-thrombotic agents was reported by 13 studies, of which 9 studies examined statistical association with outcome (**Table 3**). Trevisi et al.[37] demonstrated a significant association between use of anti-thrombotic agents and dichotomised GOS on univariate analysis (82 vs 57%; good vs poor outcome; p = 0.01), though multivariate logistic regression demonstrated no significant effect of anti-thrombotic use on dichotomised GOS (OR 0.33; 95% CI 0.10, 1.10; p = 0.02; other covariates included GCS, ASDH thickness, and MLS). Other studies demonstrated no significant association between anti-thrombotic use and GOS[25,34], dichotomised GOS[31,35], or mortality[29,34]. Examination of antiplatelet and anticoagulant use in isolation also demonstrated no significant association with GOS[30,38] or mortality[29,33].

*Mechanism and Severity of Injury*

Mechanism of injury was reported in 7 studies [21,25-27,29,33,34]. The association between mechanism of injury and outcome was examined in 5 of these studies, but a significant association was not demonstrated [21,25,27,29,33]. In all 7 studies reporting mechanism of injury, ground-level falls were the most common mechanism (range 65 – 100%), with road traffic collisions being the second most common albeit much less common than falls (range 4 – 30%).

Neurological status prior to surgical intervention was reported in all 21 included studies. Association between GCS or pupillary reactivity and outcome were examined in 11 studies. Of these, 10 studies demonstrated significant associations (**Table 3**). Neurological status was reported pre-operatively in 2 studies[21,34], on admission and pre-operatively in 2 studies[35,37], and on admission in the remaining 6 studies. Significant heterogeneity was observed in the methodology used to assess the relationship between pre-operative neurological status and outcome (**Supplementary Table 3, 4**). Statistically significant associations between GCS and GOS[30,37,38] or mortality[25,29,32-34] were demonstrated across multiple studies. Similarly, significant associations were also demonstrated between pupillary reactivity and GOS[21,25,34,35] or mortality[25,34]. No significant relationship between GCS and GOS was demonstrated in four studies[21,31,32,35], and between pupil reactivity and GOS/ mortality in four studies[29,31,37,38].

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Extracranial injuries were reported in 4 studies[20-22,28], while patients with significant extracranial injuries were excluded from five studies[19,23,24,30,36]. There were no reports of extracranial injuries in the remaining studies. None of the included studies examined the relationship between extracranial injuries and outcome. However, early studies demonstrated poor outcomes in several patients with concomitant injuries: (i) of six patients with orthopaedic fractures (three patients with femoral fractures, two patients with rib fractures and one patient with a C2 fracture), five patients died[20]; and (ii) all eight patients with orthpaedic fractures had unfavourable outcomes[21].

*Imaging*

Imaging findings with respect to ASDH were reported in 15 included studies. The association between imaging findings and outcome was examined in eight of these studies, of which five showed significant findings (**Table 3**). MLS was reported in 7 studies[25,30,31,34,35,37,38] , while ASDH thickness or volume (estimated by volumetric analysis of CT scans) was reported in 6 studies[25,30,31,35,37,38]. All studies reporting these imaging parameters analysed their association with outcome.

Petridis et al.[25] demonstrated a significantly greater proportion of patients with MLS >10mm in the GOS-1 than in the GOS-5 category (Chi-square test; p<0.005). Furthermore, the difference between ASDH thickness and MLS was quantified and its association with outcome was examined. A significant association was identified between mean ASDH-MLS difference and individual GOS groups (ANOVA; p <0.005), measuring 6.3 mm in the GOS-5 and 1.8 mm in the GOS-1 category. Similarly, three other studies demonstrated a significant association between average MLS and dichotomised GOS[30,37] or mortality[34]. One study demonstrated a negative association between MLS and GOS at both 30 days (RC -0.053, p = 0.015) and 6 months (RC -0.055, p = 0.026) on univariate linear regression, though this did not reach statistical significance on multivariate analysis[30]. Two studies demonstrated no significant association between MLS and dichotomised[38] or trichotomized[34] GOS.

Two studies demonstrated significant associations between ASDH volume and outcome[30,38]. One study demonstrated a negative association between ASDH volume and GOS at both 30

days (RC -0.006, p = 0.005) and 6 months (RC -0.005, p = 0.027) on univariate linear regression, though this did not reach statistical significance on multivariate analysis[30]. Another study demonstrated a significantly smaller ASDH volume in the favourable outcome group when compared with the unfavourable outcome group (67ml vs 118ml; unpaired Student's *t* test, p=0.05). Two studies demonstrated no significant association between ASDH thickness and GOS categories[25] or dichotomised GOS[37].

The presence of additional intracranial injuries, such as cerebral contusions and traumatic subarachnoid haemorrhage, were reported in 8 studies[20-23,25,34,36-38] but only statistically analysed with respect to outcome in 3 studies[21,25,34]. None of these studies demonstrated significant associations between additional intracranial injuries and outcome.

*Surgical factors*

Timing of surgery from the time of admission or injury was reported in 10 studies, and analysed with respect to outcome in 4 studies (**Table 3**). No statistically significant associations were demonstrated with respect to dichotomised GOS in 3 studies[21,35,38]. One study demonstrated a significant association of timing of surgery after admission, divided into 6 hour increments between <6 hours and >72 hours, and GOS (Chi-squared test; p < 0.001)[37]. However, on univariate logistic regression, timing of surgery (within 6 hours or >6 hours), demonstrated no significant effect on prediction of dichotomised GOS (p = 0.20).

Choice of surgical approach included craniotomy or DC. Four studies did not distinguish between craniotomy and decompressive craniectomy, but defined surgical intervention as the surgical evacuation of ASDH [19-22]. Of the remaining studies, 12 studies included craniotomy and DC, 4 studies included craniotomies alone, and 1 study defined surgical intervention as craniectomies [23] (**Table 1**). Number of patients undergoing each surgical approach was specifically reported in 7 studies, however, only 3 studies examined association with outcome (**Table 3**). Monsivais et al.[33] demonstrated a significant association between surgical approach (craniotomy, DC) and mortality on univariate analysis (Fisher's exact test; p<0.01), and DC as a predictor of mortality when compared with craniotomy on multivariate logistic regression (OR 5.72; 95% CI 1.11, 29.32; p = 0.04; other

covariates included age and GCS). Other studies demonstrated no significant associations between choice of surgical approach and dichotomised GOS[31,35] or mortality[35].

## Discussion

ASDH is a devastating traumatic pathology, and particularly so in the elderly. Although the significant mortality and morbidity associated with surgical intervention has been recognised for decades, there are no clear guidelines for ASDH management in this age group. With a growing elderly population worldwide and a shift in healthcare policy towards prioritising quality of life, it is essential that neurosurgical guidelines are adapted accordingly[9,10]. Indeed, the generic guidance for surgical intervention based on ASDH thickness, MLS, and neurological deterioration may not necessarily be appropriate for elderly patients with ASDH[13]. In this study, we systematically reviewed the literature for studies reporting outcomes following surgical intervention in patients aged 60 years and above with ASDH. We demonstrate that the pooled estimated mortality rate in this group is 40% at the point of discharge, and 49% at LTFU (**Figure 2**). Furthermore, the pooled estimated poor outcome (GOS 1 – 3) rate is 81% at discharge and 79% at LTFU. Analysis of separate age subgroups could not be performed due to insufficient reported data, and therefore inclusion age of above 60 years was used as a minimum criterion for consideration. Only one study included patients above 60 years of age[24], which was excluded from meta-analysis due to its date of publication. Therefore, our pooled estimated mortality and unfavourable outcome rates are generated from studies including patients aged 65 years and above. One potential source of heterogeneity was the inclusion of both craniotomy and DC within meta-analysis, as the majority of included studies did not distinguish reliably between surgical approach. Despite 11 studies including craniotomy and DC, only 7 studies reported specific numbers for each approach, and outcome could not be consistently deduced. We decided to include both procedures within the meta-analysis because: (i) we aimed to provide a pooled estimate for surgical intervention for ASDH in the elderly; and (ii) the decision between craniotomy and DC is intra-operative, therefore

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quantification of mortality and morbidity following either procedure remains pertinent. In addition, of the three studies examining effect of surgical approach on GOS or mortality, only one study demonstrated a significant association with mortality[33]. However, this was likely due to the small proportion of patients that underwent DC. Another source of heterogeneity was the duration of follow up. For the pooled estimate of mortality at LTFU, duration of follow up was 6 months in five studies, 12 months in three studies, 3 months in one study, and 2 months in one study. Similarly, for the pooled estimate of poor GOS at LTFU, duration of follow up was 6 months in four studies, 12 months in two studies, 3 months in one study, and 2 months in one study. Therefore, pooled mortality may have been underestimated given that a higher percentage of deaths would be expected at longer durations of follow up. Whilst the same concept may be true for pooled estimated poor outcomes, it is also possible that shorter durations of follow up did not capture potential future neurological recovery. Given that included studies did not consistently report the direct cause of poor outcome or mortality, it remains difficult to extrapolate any meaningful conclusions regarding this factor. Surgical inclusion or exclusion criteria were detailed in ten studies, and largely revolved around existing Brain Trauma Foundation Guidelines[13] (**Supplementary Table 2**). The possibility of comparing surgical and conservative was initially evaluated, as four studies reported outcomes in conservatively managed patients [27,32,36,37]. However, indications for conservative management were due to inoperable co-morbidities, good neurological status, moribund clinical status, or unspecified (**Supplementary Table 1**). Therefore, surgical and conservative groups were not feasible for comparison.

We also examined risk factors for mortality and poor outcome in the target cohort. Of 9 studies examining the relationship between age and outcome, only 2 studies demonstrated a statistically significant effect[30,33]. One study demonstrated a negative association between age and GOS at 30 days and 6 months on multivariate regression[30]. Whether GOS should be modelled as a continuous variable is arguable, however, as each increment does not represent an equal difference in functional status. Another study, however, demonstrated that the odds of death were almost three times higher in patients aged 80 years and above

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compared with patients aged 70 – 79 years. Whilst age appears intuitively likely to be associated with poorer outcomes, few studies demonstrate a significant association in the elderly cohort. Only two studies examined the relationship between baseline functional status and outcome. Neither demonstrated any difference between survivors and non-survivors with respect to pre-morbid independence[29,33] or pre-admission residence[29]. Further studies are required to elucidate this relationship, since the use of scoring systems such as Karnofsky Performance Status[40] or Clinical Frailty Score[41] may provide a feasible method to prognosticate surgical candidates. This has been demonstrated by a recent study, which highlighted the use of the Modified Frailty Index[42] to predict 30 day mortality and 6 month unfavourable outcome[43]. Similarly, four studies reported the relationship between co-morbidities and outcome, though only one study used a formal scoring system for co-morbidities[37]. However, no significant association with dichotomised GOS was demonstrated[37]. In a similar fashion to baseline functional status, further studies using scoring systems such as the Charlston Co-morbidity Index[44] would be useful for assessment for neurosurgical decision-making.

In contrast, the association between neurological status and outcome were examined in 11 studies. Although significant associations were demonstrated across several studies, a considerable degree of heterogeneity was observed in the approach to analysis (**Supplementary Table 3, 4**). GCS was modelled as a continuous variable, dichotomised at various points, or categorised with respect to severity of TBI. However, the emerging theme was that a lower GCS on admission was associated with a greater risk of mortality. Some studies demonstrated a similar relationship with risk of poor GOS[30,37,38], though this did not consistently reach statistically significance[21,31,32,35]. Therefore, identifying a cut-off for admission GCS could be an important avenue for future research to help identify patients that are highly unlikely to benefit from surgery. Pupillary reactivity to light was also examined across several studies. As expected, fixed and dilated pupils were often associated with poor outcomes and mortality (**Supplementary Table 4**), although some studies revealed no significant association. This could be due to subjective errors on reporting of

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pupillary reactivity, which may be remedied in future studies with use of pupillometers. In the absence of a reliable cut-off for pre-operative GCS, understanding the relationship between pupillary reactivity and outcome is essential for prognosticating intubated patients. Several studies examined the relationship between imaging findings and outcome, demonstrating a significant association between MLS and outcome in four studies[25,30,34,37]. Similar to the assessment of pre-operative neurology, approaches to analysis were varied, with no reliable conclusion with respect to a prognostic cut-off value for MLS. Interestingly, whilst ASDH thickness was not significantly associated with outcome [25,37], ASDH volume was significantly associated with GOS in two studies[30,38]. The reported negative association between ASDH-MLS differential and GOS may also warrant further investigation[25]. This has been previously reported in a retrospective cohort study of ASDH patients, which demonstrated a 75% mortality rate in patients with an ASDH-MLS differential of >5mm[45]. In contrast, the group with an ASDH-MLS differential of 0mm only had a 25% mortality rate. Therefore, further studies comparing different measurement approaches may yield useful information regarding prognostication.

Of note, two key studies[46,47] could not be included due to failure to meet our eligibility criteria. Shibahashi et al.[46] compared outcomes following craniotomy and DC for ASDH across 1028 patients matched for age, gender, admission GCS and vital signs, and comorbidities. Interestingly, no differences in mortality between treatment groups in the elderly cohort were found. Indeed, this finding supports our approach to a pooled estimate for mortality on meta-analysis. This study was not included in our review due to it not presenting absolute mortality figures, rather a mean difference between groups only. While it provides some evidence of no difference in mortality when comparing craniotomy with DC, the groups were not matched for pupillary abnormalities, nor were any imaging parameters taken into account in the matching process. One recent study was published outside of the dates of our search strategy, examining a total of 2508 patients aged 65 years and older that underwent surgical evacuation of ASDH [47]. Data was collected from the USA National Trauma Data Bank registry between 2016 and 2017, which spans approximately 765 centres. Interestingly, inpatient mortality was estimated at only 30.5%. Multivariate

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logistic regression revealed that age, pre-operative GCS, traumatic subarachnoid haemorrhage, and MLS > 5 mm were significant predictors of mortality. Although the risk factors are in keeping with findings from previous studies, the reported inpatient mortality was comparatively low. Given that surgical approach was not specified, as ICD-10 diagnosis codes were used to extract data, and surgical indications may have varied significantly between centres, it is unclear whether inpatient mortality is truly representative of this population.

## Conclusion

In this systematic review and meta-analysis, we demonstrate that outcomes following surgical evacuation of ASDH in patients aged 65 years and above are poor. Pooled estimated mortality rates in this group are 40% at discharge and 49% at LTFU. Estimated rates of poor GOS are 81% at discharge and 79% at LTFU. Potential risk factors for poor outcome include age, baseline functional status, pre-operative neurological status, and imaging parameters. Whilst further studies are required to reliably characterise predictors of poor outcome to inform separate guidelines for surgical intervention in this cohort, this systematic review constitutes the best level of evidence in the current literature that surgical intervention for ASDH in the elderly carries significant risks, which must be weighed against benefits. We, herein, provide robust evidence that helps healthcare professional decision-making and counselling of families on a contentious issue.

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## Contributions

**SM**- data curation, formal analysis, project administration, software, writing- original draft; **RS**- data curation, formal analysis, project administration, software, writing- original draft; **OM**- investigation, methodology; **IM**- investigation, methodology; **OE**- investigation, methodology; **JM**- investigation, methodology, writing- reviewing & editing; **MZ**- conceptualization, supervision, writing- review & editing

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**Competing Interests Statement**

Authors have no competing interests to declare.

**Data Availability**

Available on request to corresponding author.

**Patient and Public involvement**

There was no patient or public involvement in this research study.

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**Table 1.** Summary of included studies of ASDH in elderly following systematic review of the literature. Abbreviations: DC- decompressive craniectomy, GCS- Glasgow Coma Score, GOS- Glasgow Outcome Scale, ICP- intracranial pressure, SDH- subdural haematoma.

| Author, Year | Location (Period) | Country | No. of hospitals/ Patients | Age for inclusion (years) | Age (years) % Male | Treatment | Outcomes measured | Risk factor analysis | STROBE criteria |
|---|---|---|---|---|---|---|---|---|---|
| Wilberger et al., 1991[19] | Allegheny General Hospital (1982 to 1987) | USA | 1/ 28 | > 65 | N/A | Craniotomy | Mortality, GOS at discharge | Age, ICP, GCS, timing of surgery; descriptive analyses (not in elderly group alone) | 17 |
| Cagetti et al., 1992[20] | University of Genoa Medical School (Jan 1980 to Dec 1988) | Italy | 1/ 26 | > 80 | 84.2 (mean) 57.7% | Surgical evacuation | Mortality, GOS at discharge | Age, mechanism, GCS, comorbidities; descriptive analyses (not in elderly group alone) | 10 |
| Jamjoom, 1992[21] | Frenchay Hospital (1980 to 1989) | UK | 1/ 27 | >= 75 | 79.2 (mean) 55.6% | Surgical evacuation | Mortality, GOS at 6 months | Age, sex, mechanism, GCS/ pupils, imaging, timing of surgery; Chi-squared test | 14 |
| Kotwica & Jakubowski, 1992[22] | Medical University of Lodz (1984 to 1990) | Poland | 1/ 27 | > 70 | N/A | Surgical evacuation | Mortality, GOS at discharge | Age, gender, GCS, imaging; descriptive analyses | 15 |

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|---|---|---|---|---|---|---|---|---|---|
| Massaro et al., 1996[23] | Centro Traumatologico Ortopedico (1982 to 1992) | Italy | 1/ 25 | > 65 | N/A | DC | Mortality, GOS at 18 months | Age, gender, mechanism, GCS, timing of surgery, imaging; Chi-squared test/ Fisher's exact test (not in elderly group alone) | 15 |
| Koc et al., 1997[24] | Erciyes University (Jan 1986 to Aug 1995) | Turkey | 1/ 15 | >= 61 | N/A | Craniotomy | Mortality, GOS at 3 months | Age, gender, presenting GCS; Chi-squared test (not in elderly group alone) | 10 |
| Petridis et al., 2009[25] | UKSH Campus Kiel | Germany | 1/ 119 | > 65 | N/A<br>55.2% | Craniotomy, DC | Mortality, GOS discharge | GCS/ pupils, imaging, anticoagulation, ICP; Chi-squared test, ANOVA | 13 |
| Hanif et al., 2009[26] | Beaumont Hospital (Jan 1999 to Dec 2003) | Ireland | 1/ 29 | > 70 | 77.0 (mean)<br>59.7% | Surgical evacuation | Mortality, GOS at 6 months | Age, GCS; descriptive analyses (not in elderly group alone) | 13 |
| Taussky et al., 2012[27] | Kontansspital Aarau (Jan 2002 to Dec 2007) | Switzerland | 1/ 37 | > 65 | 73.0 (median)<br>46% | Craniotomy - 23<br>DC - 14 | Mortality at discharge<br>Mortality, GOS at 6 months | Age, pupils, timing of surgery, GCS/ pupils, anticoagulants, co-morbidities; descriptive analyses | 20 |

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|---|---|---|---|---|---|---|---|---|---|
| Hamed et al., 2016[39] | Rheinische Friedrich-Wilhelms University (Jan 2010 to Dec 2014) | Germany | 1/57 | >70 | N/A | Craniotomy, DC | GOS at 6 months | Age, TBI severity, timing of admission, surgical approach, anti-thrombotics, timing of surgery, duration of hospital stay, post-operative complications (not for elderly group alone) | 17 |
| Merzo et al., 2016[28] | Uppsala University Hospital (2008 to 2010) | Sweden | 1/ 24 | >= 65 | N/A | Craniotomy – 23 DC – 1 | Mortality, GOS at 6 months | Age, gender, comorbidities, mechanism, imaging; Chi-squared, unpaired Student's t-test (not for elderly group alone) | 17 |
| Raj et al., 2016[29] | Helsinki University Hospital (Jan 2009 to Dec 2012) | Finland | 1/ 44 | >= 75 | 81.0 (median) 52% | Craniotomy, DC | Mortality at 1 and 3 years | Age, baseline functional status, residence, gender, anticoagulants, mechanism, GCS/ pupils; Chi-squared test, unpaired Student's t test, Mann Whitney U test | 18 |
| Benedetto et al., 2017[30] | Azienda Ospedaliero-Universitaria Pisana (June 2011 to Dec 2014) | Italy | 1/ 67 | > 70 | 80.5 (median) 53.7% | Craniotomy | Mortality, GOS at 1 and 6 months | Age, imaging, GCS, antiplatelets; simple and multiple linear regression | 17 |

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|---|---|---|---|---|---|---|---|---|---|
| McGinity et al., 2017[31] | University of Texas Health Science Center (2005 to 2015) | USA | 1/ 34 | >= 80 | 84.0 (mean)<br>N/A | Craniotomy – 32<br>DC – 2 | Mortality, GOS at 2 months (variable) | Age, mechanism, type of surgery, anticoagulants, GCS/ pupils, imaging, comorbidities; Fisher's exact test, Mann Whitney U test | 17 |
| Won et al., 2017[32] | Goethe-University Hospital (Jan 2007 to Dec 2016) | Germany | 1/ 56 | >= 80 | 85.0 (mean)<br>43% | Craniotomy, DC | Mortality at discharge | GCS; univariate/ multivariate logistic regression | 20 |
| Monsivais et al., 2018[33] | University of Texas Health Science Centre (Jan 2006 to Jul 2016) | USA | 1/ 119 | >= 70 | N/A | Craniotomy – 111<br>DC – 18 | Mortality at discharge | Age, GCS, type of surgery; Chi-squared test/ Fisher's exact test | 20 |
| Akbik et al., 2019[34] | University of New Mexico Hospital (Jan 2013 to Dec 2017) | USA | 1/ 62 | >= 65 | 78.0 (median)<br>48% | Craniotomy, DC | Mortality, GOS at discharge and 3 months | Age, comorbidities, GCS/ pupils, imaging, antithrombotics; Kruskal Wallis, Fisher's exact test | 15 |

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|---|---|---|---|---|---|---|---|---|---|
| Bus et al., 2019[35] | Academic Medical Centre (Jan 2000 to Oct 2015) | Netherlands | 1/ 84 | >= 65 | 75.0 (mean) 64.3% | Craniotomy – 74 DC – 10 | Mortality at discharge Mortality, GOS at 1 year | Age, gender, imaging, GCS/ pupils, age of SDH, imaging, timing of surgery, anticoagulation; Chi-squared/ Fisher's exact tests, Mann Whitney U test, univariate logistic regression | 17 |
| Sufaro et al., 2019[36] | Soroka University Medical Center (2006 to 2016) | Israel | 1/ 28 | > 70 | 81.5 (mean) 53.6% | Craniotomy | Mortality, mRS score at discharge and 1 year | Age, gender, baseline status, antithrombotics, GCS, imaging, neurological deficit; descriptive analyses | 17 |
| Trevisi et al., 2020[37] | Ospedale Santo Spirito; Fondazione Policlinico Universitario A. Gemelli IRCSS; S. Anna University Hospital, UOC Neurochirurgia (Jan 2016 to Dec 2019) | Italy | 5/ 147 | >= 70 | 78 (mean) 58% | Craniotomy- 133 DC- 14 | Mortality, GOS at discharge and 6 months | Age, gender, comorbidities, GCS/ pupils, imaging, side of SDH, size of craniotomy, timing of surgery, anticoagulants; Chi-squared test, univariate/ multivariate logistic regression | 16 |

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|---|---|---|---|---|---|---|---|---|---|
| Younsi et al., 2020[38] | University Hospital Heidelburg (Jan 2006 to Dec 2016) | Germany | 1/ 27 | >= 80 | 84.0 (median) 60% | Craniotomy, DC | Mortality, GOS at discharge | Age, gender, co-morbidities, GCS/ pupils, imaging, surgical factors, anticoagulants; Fisher's exact test, unpaired Student t test | 21 |

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**Table 2.** Mortality, outcomes, and duration of follow up following surgical evacuation of ASDH in patients aged 60 years and above.

Abbreviations: GOS- Glasgow Outcome Scale.

| Author, Year | No. of Patients | *At Discharge* (N, %) | | | *At Long Term Follow Up* (N, %) | | | |
|---|---|---|---|---|---|---|---|---|
| | | Deaths | GOS 1 – 3 | GOS 4 – 5 | Deaths | GOS 1 – 3 | GOS 4 – 5 | Duration |
| Wilberger et al., 1991[19] | 28 | 23 | - | - | - | - | - | - |
| Cagetti et al., 1992[20] | 26 | 23 | 23 | 3 | - | - | - | - |
| Kotwica & Jakubowski, 1992[22] | 27 | 23 | - | - | - | - | - | - |
| Jamjoom, 1992[21] | 27 | - | - | - | 19 | 23 | 4 | 6 |
| Massaro et al., 1996[23] | 25 | - | - | - | 15 | 22 | 3 | 18 |
| Koc et al., 1997[24] | 15 | - | - | - | 11 | 12 | 3 | 3 |
| Petridis et al., 2009[25] | 119 | 64 | 94 | 25 | - | - | - | - |
| Taussky et al., 2012[27] | 37 | 13 | - | - | 13 | 22 | 15 | 6 |
| Raj et al., 2015[29] | 44 | - | - | - | 22 | - | - | 12 |
| Merzo et al., 2016[28] | 24 | - | - | - | 9 | 19 | 5 | 6 |
| Hamed et al., 2016[39] | 57 | - | - | - | - | 53 | 4 | 6 |
| Benedetto et al., 2017[30] | 67 | 37 | 60 | 7 | 45 | 58 | 9 | 6 |
| Won et al., 2017[32] | 56 | 13 | - | - | - | - | - | - |

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|---|---|---|---|---|---|---|---|---|
| McGinity et al., 2017[31] | 33 | - | - | - | 18 | 27 | 6 | 2 |
| Monsivais et al., 2018[33] | 112 | 47 | - | - | - | - | - | - |
| Akbik et al., 2019[34] | 62 | 24 | 54 | 8 | 27 | 45 | 17 | 3 |
| Bus et al., 2019[35] | 84 | 44 | - | - | 54 | 63 | 21 | 12 |
| Sufaro et al., 2019[36] | 28 | 6 | - | - | 12 | - | - | 12 |
| Trevisi et al., 2020[37] | 147 | 51 | 114 | 33 | 64 | 115 | 32 | 6 |
| Younsi et al., 2020[38] | 27 | 9 | 21 | 6 | - | - | - | - |

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**Table 3.** Potential risk factors for mortality or poor outcome following surgical evacuation of ASDH in patients aged 60 years and above. N- not reported; X- reported but not statistically analysed; NS- non-significant association on statistical analysis; S- significant association on statistical analysis. Abbreviations: DC- decompressive craniectomy, GCS- Glasgow Coma Score, GOS- Glasgow Outcome Scale.

| Author, Year | Outcome | Intervention | Risk Factors | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Age | Sex | Base-line | GCS/ pupils | Imaging | Mechanism of Injury | Other intracranial injuries | Timing of surgery | Craniotomy vs DC | Antithrombotic agents |
| Wilberger et al., 1991[19] | Mortality GOS | Surgical | X | X | N | X | N | N | N | X | N | N |
| Cagetti et al., 1992[20] | Mortality GOS | Surgical | X | X | X | X | N | N | X | N | N | N |
| Jamjoom, 1992[21] | GOS | Surgical | NS | X | N | S | NS | NS | NS | NS | N | N |
| Kotwica & Jakubowski, 1992[22] | GOS | Surgical | X | X | N | X | X | N | X | N | N | N |
| Massaro et al., 1996[23] | Mortality GOS | DC | X | X | N | X | X | N | X | X | N | N |
| Koc et al., 1997[24] | Mortality GOS | Craniotomy | X | X | N | X | X | N | N | N | N | N |
| Hanif et al., 2009[26] | Mortality GOS | Craniotomy | X | X | N | X | N | X | N | N | N | N |
| Petridis et al., 2009[25] | Mortality GOS | Craniotomy, DC | X | X | N | S | S | NS | NS | N | NS | NS |
| Taussky et al., 2012[27] | GOS | Craniotomy, DC | X | X | X | X | N | NS | N | X | X | X |
| Hamed et al., 2016[39] | GOS | Craniotomy, DC | X | N | N | X | X | N | N | X | X | X |

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| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Merzo et al., 2016[28] | Mortality GOS | Craniotomy, DC | X | N | X | X | N | N | N | N | X | N |
| Raj et al., 2016[29] | Mortality | Craniotomy, DC | NS | NS | NS | S | N | NS | N | N | N | NS |
| Benedetto et al., 2017[30] | GOS | Craniotomy | S | X | X | S | S | N | N | N | X | NS |
| McGinity et al., 2017[31] | GOS | Craniotomy, DOC | NS | N | NS | NS | NS | N | N | N | NS | NS |
| Won et al., 2017[32] | Mortality GOS | Craniotomy, DC | X | X | X | S | X | N | N | X | N | X |
| Monsivais et al., 2018[33] | Mortality | Craniotomy, DC | S | N | NS | S | N | NS | N | X | S | NS |
| Akbik et al., 2019[34] | Mortality GOS | Craniotomy, DC | NS | X | N | S | S | X | NS | N | N | NS |
| Bus et al., 2019[35] | Mortality GOS | Craniotomy, DC | NS | NS | N | S | NS | N | N | NS | NS | NS |
| Sufaro et al., 2019[36] | Mortality mRS | Craniotomy | X | X | X | X | X | N | X | N | N | X |
| Trevisi et al., 2020[37] | GOS | Craniotomy, DC | NS | NS | NS | S | S | N | X | S | X | S |
| Younsi et al., 2020[38] | GOS | Craniotomy, DC | NS | NS | S | S | S | N | X | NS | N | NS |

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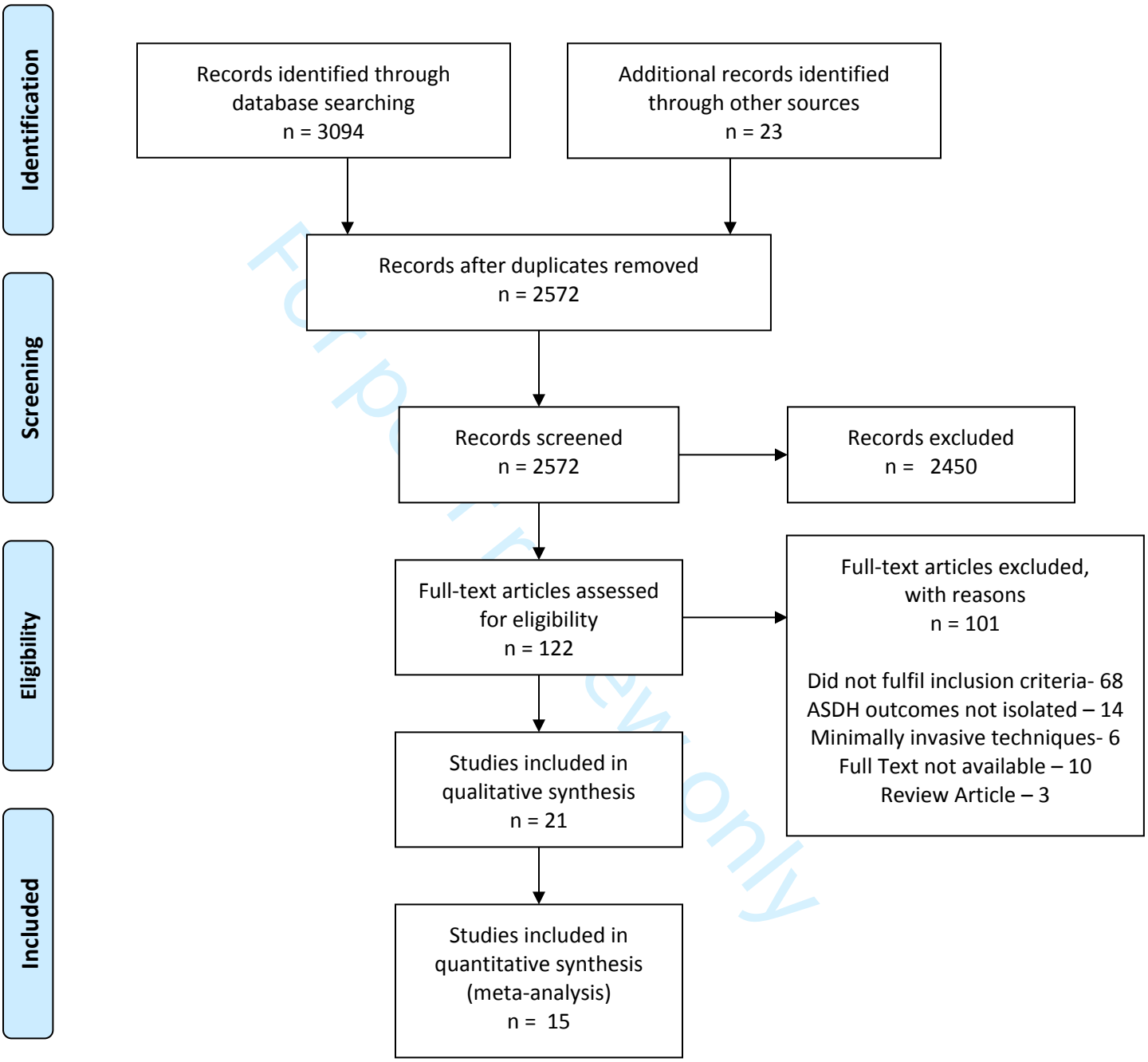
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## Figure Legends

**Figure 1.** Search strategy for systematic literature review in accordance with PRISMA guidelines.

**FIgure 2.** Forest plots depicting pooled estimates of mortality at discharge (A), poor outcome at discharge (B), mortality at discharge (C), and mortality at long term follow up (LTFU) (D) in patients aged 65 years and above that underwent surgical evacuation of ASDH.

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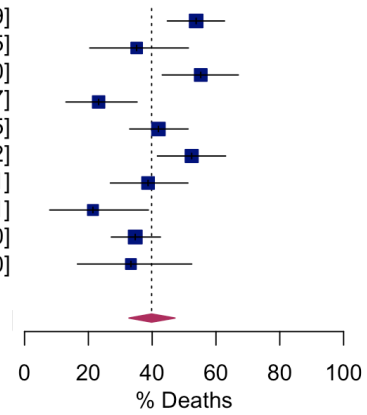

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**A**

## Deaths at discharge

| Study | Deaths | Total | % Deaths | 95% C.I. |
|---|---|---|---|---|
| Petridis 2009 | 64 | 119 | 53.78 | [44.75; 62.69] |
| Taussky 2012 | 13 | 37 | 35.14 | [20.44; 51.35] |
| Benedetto 2017 | 37 | 67 | 55.22 | [43.15; 67.00] |
| Won 2017 | 13 | 56 | 23.21 | [12.97; 35.27] |
| Monsivais 2018 | 47 | 112 | 41.96 | [32.95; 51.25] |
| Bus 2018 | 44 | 84 | 52.38 | [41.63; 63.02] |
| Akbik 2019 | 24 | 62 | 38.71 | [26.90; 51.21] |
| Sufaro 2019 | 6 | 28 | 21.43 | [ 7.89; 38.81] |
| Trevisi 2020 | 51 | 147 | 34.69 | [27.19; 42.60] |
| Younsi 2020 | 9 | 27 | 33.33 | [16.58; 52.40] |
| **Total** | **308** | **739** | | |
| **Random effects model** | | | **39.83** | **[32.73; 47.14]** |

Heterogeneity: $I^2$ = 73%, $\tau^2$ = 0.0095, $\chi^2_9$ = 33.75 ($p$ < 0.01)

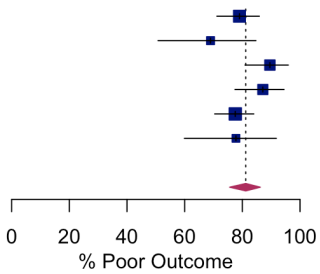



**B**

## Poor outcome at discharge

| Study | Poor Outcome | Total | % Poor Outcome | 95% C.I. |
|---|---|---|---|---|
| Petridis 2009 | 94 | 119 | 78.99 | [71.17; 85.88] |
| Hanif 2009 | 20 | 29 | 68.97 | [50.77; 84.71] |
| Benedetto 2017 | 60 | 67 | 89.55 | [80.93; 95.92] |
| Akbik 2019 | 54 | 62 | 87.10 | [77.45; 94.46] |
| Trevisi 2020 | 114 | 147 | 77.55 | [70.42; 83.96] |
| Younsi 2020 | 21 | 27 | 77.78 | [59.90; 91.79] |
| **Total** | **363** | **451** | | |
| **Random effects model** | | | **81.18** | **[75.61; 86.21]** |

Heterogeneity: $I^2$ = 45%, $\tau^2$ = 0.0029, $\chi^2_5$ = 9.12 ($p$ = 0.10)

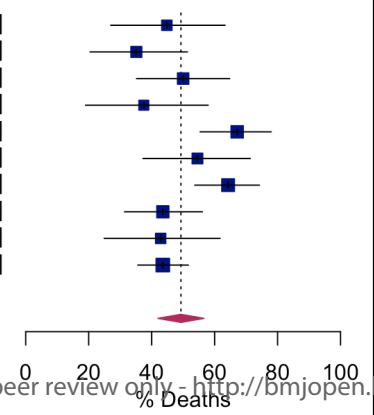



**C**

## Deaths at Long Term Follow Up

| Study | Deaths | Total | % Deaths | 95% C.I. |
|---|---|---|---|---|
| Hanif 2009 | 13 | 29 | 44.83 | [27.00; 63.32] |
| Taussky 2012 | 13 | 37 | 35.14 | [20.44; 51.35] |
| Raj 2015 | 22 | 44 | 50.00 | [35.19; 64.81] |
| Merzo 2016 | 9 | 24 | 37.50 | [19.00; 57.95] |
| Benedetto 2017 | 45 | 67 | 67.16 | [55.39; 77.96] |
| McGinity 2017 | 18 | 33 | 54.55 | [37.25; 71.32] |
| Bus 2018 | 54 | 84 | 64.29 | [53.69; 74.23] |
| Akbik 2019 | 27 | 62 | 43.55 | [31.38; 56.11] |
| Sufaro 2019 | 12 | 28 | 42.86 | [24.94; 61.73] |
| Trevisi 2020 | 64 | 147 | 43.54 | [35.60; 51.64] |
| **Total** | **277** | **555** | | |
| **Random effects model** | | | **49.30** | **[42.01; 56.61]** |

Heterogeneity: $I^2$ = 63%, $\tau^2$ = 0.0079, $\chi^2_9$ = 24.20 ($p$ < 0.01)

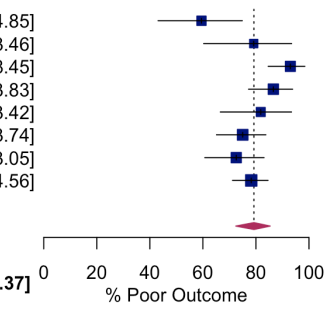



**D**

## Poor outcome at Long Term Follow Up

| Study | Poor Outcome | Total | % Poor Outcome | 95% C.I. |
|---|---|---|---|---|
| Taussky 2012 | 22 | 37 | 59.46 | [43.11; 74.85] |
| Merzo 2016 | 19 | 24 | 79.17 | [60.30; 93.46] |
| Hamed 2016 | 53 | 57 | 92.98 | [84.62; 98.45] |
| Benedetto 2017 | 58 | 67 | 86.57 | [77.22; 93.83] |
| McGinity 2017 | 27 | 33 | 81.82 | [66.57; 93.42] |
| Bus 2018 | 63 | 84 | 75.00 | [65.13; 83.74] |
| Akbik 2019 | 45 | 62 | 72.58 | [60.72; 83.05] |
| Trevisi 2020 | 115 | 147 | 78.23 | [71.17; 84.56] |
| **Total** | **402** | **511** | | |
| **Random effects model** | | | **79.25** | **[72.42; 85.37]** |

Heterogeneity: $I^2$ = 66%, $\tau^2$ = 0.0079, $\chi^2_7$ = 20.65 ($p$ < 0.01)




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**Supplementary Table 1.** Indications for conservative management in included studies, and subsequent mortality/ outcomes.

| Author, Year | No. of Patients | *At Discharge* (N, %) | | | *At Long Term Follow Up* (N, %) | | | | Indications |
|---|---|---|---|---|---|---|---|---|---|
| | | Deaths | GOS 1 – 3 | GOS 4 – 5 | Deaths | GOS 1 – 3 | GOS 4 – 5 | Duration | |
| Taussky et al., 2012[1] | 5 | - | - | - | - | 5 | 0 | 6 | Any patients not fulfilling criteria for surgery (see **Supplementary Table 2**), or with bilaterally fixed pupils. |
| Won et al., 2017[2] | 7 | - | - | - | 5 | 7 | 0 | 3 | Not specified. Seven out of 68 patients were managed conservatively due to inoperable comorbidities. |
| Sufaro et al., 2019[3] | 26 | 2 | - | - | 9 | - | - | 12 | No specific indications for conservative management given. No significant differences between surgical and conservative groups except for ASDH thickness and focal neurological deficit. |
| Trevisi et al., 2020[4] | 66 | 14 | 21 | 45 | 21 | 27 | 39 | 6 | No specific indications for conservative management. Significantly larger proportion of conservatively managed patients in GCS 13 - 15 category (77%) compared with surgical (36%) group. |

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**Supplementary Table 2.** Indications for surgical intervention in included studies.

| Author, Year | Indication/ Prerequisites |
|---|---|
| Wilberger et al., 1991[5] | Surgical indication not specified.<br>All included patients had ASDH thickness >= 3mm and MLS >5 mm.<br>Following patients were excluded from analyses:<br>● Timing of surgery could not be determined<br>● Open head injuries, prolonged hypotension/ hypoxia, severe life-threatening extra-cranial injuries |
| Massaro et al., 1996[6] | Surgical indications- MLS > 5 mm<br>Following patients were excluded- open head injury, prolonged hypotension, hypoxia or severe life threatening injuries |
| Koc et al., 1997[7] | Surgical indications- ASDH thickness > 10mm, MLS >5 mm |
| Taussky et al., 2012[1] | Following patients considered suitable for surgical intervention:<br>● Pre-morbid functional status- KPS of at least 80, usually independent<br>● Co-morbidities- no evidence of dementia, no comorbidities limiting survival to less than 12 months<br>● Surgical indications- MLS =>5mm; GCS =<13 |
| Merzo et al., 2016[8] | Not specified for ASDH patients, though general indications for surgical intervention in TBI patients given |
| Raj et al., 2016[9] | Brain trauma foundation guidelines |
| Benedetto et al., 2017[10] | Following patients considered suitable for surgical intervention:<br>● ASDH thickness > 10 mm or MLS >5 mm with patient of any GCS<br>● GCS drop of 2 points or more, with less severe ASDH thickness/ MLS<br>Following patients were excluded from analysis:<br>● Bilateral fixed pupils<br>● Concomitant EDH or significant cerebral contusions<br>● Major thoracic or abdominal trauma |
| Monsivais et al., 2018[11] | Following patients considered unsuitable for surgical intervention:<br>● Neurologically poor- GCS 3 - 5 with or without pupillary involvement with evidence of impending herniation on CT imaging<br>● Haemodynamic instability, unsuitable for ventilation, or severe cardiac/ pulmonary compromise |

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| | |
|---|---|
| | • Advanced malignancy with metastatic disease |
| Sufaro et al., 2019[3] | Following patients considered suitable for inclusion:<br>• GCS 13 - 15 and at least one of the following- ASDH thickness > 10mm, MLS > 5 mm, GCS drop of 2 points or more from time of injury<br>• Surgical intervention performed predominantly in patients with evolving hemiparesis<br>Patients with significant major injuries were excluded. |
| Trevisi et al., 2020[4] | Following patients considered suitable for surgical intervention:<br>• ASDH thickness > 10mm, MLS >5 mm (unless other clinical features/ demographics/ baseline functional status were not in favour of surgical intervention- objective parameters not specified) |
| Cagetti et al., 1992[12] | Not specified |
| Kotwica & Jakubowski, 1992[13] | Not specified |
| Akbik et al., 2009[14] | Not specified |
| Petridis et al., 2009[15] | Not specified |
| Hamed et al., 2016[16] | Not specified |
| Won et al., 2017[2] | Not specified |
| Bus et al., 2019[17] | Not specified |
| McGinity et al., 2017[18] | Not specified |
| Younsi et al., 2020[19] | Not specified |

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**Supplementary Table 3.** Association between pre-operative neurological status and mortality/ poor outcome following surgical evacuation of ASDH in patients aged 60 years and above.

| Study | GCS | Outcome | Statistical Test |
|---|---|---|---|
| Jamjoom, 1992[20] | Dichotomised: >=5 vs <5 | Dichotomised GOS | Chi-squared test; NS |
| Petridis et al., 2009[15] | Categorised: 13-15, 9-12, 3-8 | Mortality | Chi-squared test; p < 0.001 |
| Raj et al., 2016[9] | Categorised: 13-15, 9-12, 3-8 | Mortality | Chi-squared test; p < 0.001 |
| Benedetto et al., 2017[10] | GCS | GOS (30 days) | Univariate linear regression; RC 0.18, p<0.0001 |
| | | | Multivariate linear regression; RC 0.17, p<0.0001 |
| | | GOS (6 months) | Univariate linear regression; RC 0.20, p<0.0001 |
| | | | Multivariate linear regression; RC 0.20, p<0.0001 |
| Won et al., 2017[2] | Dichotomised: >6 vs <=6 | Mortality | Univariate logistic regression; OR 4.0, p = 0.04 |
| | | Dichotomised GOS (discharge) | Univariate logistic regression; OR 4.2, NS |
| | | Dichotomised GOS (LTFU) | Univariate logistic regression; OR 3.7, NS |
| Monsivais et al., 2018[11] | Dichotomised: >=9 vs <9 | Mortality | Chi-squared test; p = 0.01<br>Multivariate logistic regression; OR 3.0, p = 0.02 |

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| | | | |
|---|---|---|---|
| Akbik et al., 2019[14] | GCS (mean) | Mortality | Kruskal Wallis test; p = 0.014 |
| | | Categorised GOS | Kruskal Wallis test; p = 0.016 |
| Bus et al., 2019[17] | Categorised: 3-8, 9-15 | Dichotomised GOS | Chi-squared test; NS |
| Trevisi et al., 2020[4] | GCS | Dichotomised GOS | Univariate logistic regression; p < 0.001 |
| | | | Multivariate logistic regression; OR 0.87, p = 0.04 |
| | Dichotomised: >8 vs <=8 | Dichotomised GOS | Chi-squared test; p < 0.001 |

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**Supplementary Table 4.** Association between pre-operative pupil reactivity to light and mortality/ poor outcome following surgical evacuation of ASDH in patients aged 60 years and above.

| Study | Pupils | Outcome | Statistical Test |
|---|---|---|---|
| Jamjoom, 1992[20] | Non-reactive pupil(s) | Dichotomised GOS | Chi-squared test; p = 0.025 |
| Petridis et al., 2009[15] | Bilaterally reactive, unilaterally reactive, bilaterally unreactive | Mortality<br>GOS (1 vs others) | Chi-squared test; p<0.001 |
| Raj et al., 2016[9] | Normal, abnormal | Mortality | Chi-squared test; NS |
| Akbik et al., 2019[14] | Fixed and dilated, normal | Mortality | Fisher's exact test; p = 0.021 |
| | | Categorised GOS | Fisher's exact test; p = 0.009 |
| Bus et al., 2019[17] | Bilaterally dilated and unreactive | Dichotomised GOS | Fisher's exact test; p = 0.03 |
| Trevisi et al., 2020[4] | Fixed pupils | Dichotomised GOS | Chi-squared test; NS |
| Younsi et al., 2020[19] | Anisocoria | Dichotomised GOS | Fisher's exact test; NS |

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**The PRISMA for Abstracts Checklist**

| TITLE | CHECKLIST ITEM | REPORTED ON PAGE # |
|---|---|---|
| 1. Title: | Identify the report as a systematic review, meta-analysis, or both. | 1 |
| **BACKGROUND** | | |
| 2. Objectives: | The research question including components such as participants, interventions, comparators, and outcomes. | 4 |
| **METHODS** | | |
| 3. Eligibility criteria: | Study and report characteristics used as criteria for inclusion. | 5 |
| 4. Information sources: | Key databases searched and search dates. | 4 |
| 5. Risk of bias: | Methods of assessing risk of bias. | 6 |
| **RESULTS** | | |
| 6. Included studies: | Number and type of included studies and participants and relevant characteristics of studies. | 7 |
| 7. Synthesis of results: | Results for main outcomes (benefits and harms), preferably indicating the number of studies and participants for each. If meta-analysis was done, include summary measures and confidence intervals. | 7 |
| 8. Description of the effect: | Direction of the effect (i.e. which group is favoured) and size of the effect in terms meaningful to clinicians and patients. | 7 |
| **DISCUSSION** | | |
| 9. Strengths and Limitations of evidence: | Brief summary of strengths and limitations of evidence (e.g. inconsistency, imprecision, indirectness, or risk of bias, other supporting or conflicting evidence) | 12 |
| 10. Interpretation: | General interpretation of the results and important implications | 12-13 |
| **OTHER** | | |
| 11. Funding: | Primary source of funding for the review. | N/A |
| 12. Registration: | Registration number and registry name. | 2 |

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# Acute subdural haematoma in the elderly- to operate or not to operate? A systematic review and meta-analysis of outcomes following surgery



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# Acute subdural haematoma in the elderly- to operate or not to operate? A systematic review and meta-analysis of outcomes following surgery

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| | |
|---|---|
| Abstract | 298 |
| Text | 4619 |
| References | 48 |
| Tables | 3 |
| Figures | 2 |
| Supplementary Data | 7 |

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## Abstract

**Objectives**: Acute subdural haematoma (ASDH) is a devastating pathology commonly found on CT brain scans of traumatic brain injury (TBI) patients. The role of surgical intervention in the elderly has been increasingly questioned due to its associated morbidity and mortality. Therefore, a systematic review and meta-analysis of the literature to quantify the mortality and functional outcomes associated with surgical management of ASDH in the elderly was performed.

**Design/ Setting**: A multi-database literature search between January 1990 and May 2020, and meta-analysis of proportions was performed to quantify mortality and unfavourable outcome (GOS 1 – 3; death/ severe disability) rates.

**Participants**: Studies reporting patients aged 60 years or older.

**Interventions**: craniotomy, decompressive craniectomy, conservative management.

**Outcome Measures**: mortality and functional outcomes (discharge, long term follow up (LTFU)).

**Results**: 2572 articles were screened, yielding 21 studies for final inclusion and 15 for meta-analysis. Pooled estimates of mortality were 39.83% (95% CI: 32.73, 47.14%; 10 studies, 308/739 patients, $I^2$=73%) at discharge and 49.30% (95% CI: 42.01, 56.61%; 10 studies, 277/555 patients, $I^2$=63%) at LTFU. Mean duration of follow up was 7.1 months (range 2 – 12 months). Pooled estimate of percentage of poor outcomes was 81.18% (95% CI: 75.61, 86.21%; 6 studies, 363/451 patients, $I^2$=45%) at discharge, and 79.25% (95% CI: 72.42, 85.37%; 8 studies, 402/511 patients, $I^2$=66%) at LTFU. Mean duration of follow up was 6.4 months (range 2 – 12 months). Potential risk factors for poor outcome included age, baseline functional status, pre-operative neurological status, and imaging parameters.

**Conclusions**: Outcomes following surgical evacuation of ASDH in patients aged 60 years and above are poor. This constitutes the best level of evidence in the current literature that surgical intervention for ASDH in the elderly carries significant risks, which must be weighed against benefits.

**Study registration**: Study protocol is registered on PROSPERO (CRD42020189508).



**Keywords**

Traumatic brain injury, elderly, acute subdural haematoma, craniotomy


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**Strengths and limitations of this study**

- Critical appraisal of studies reporting both mortality and functional outcomes following surgical evacuation of ASDH in the elderly
- Examination of risk factors for poor outcome in included studies
- Heterogeneity arising from potential differences in indications for surgical intervention
- Potential reporting bias and heterogeneity in meta-analysis from inclusion of smaller retrospective studies

## Introduction

Traumatic brain injury (TBI) is amongst the leading causes of mortality and morbidity worldwide[1,2]. We are currently experiencing an increasing contribution of traumatic injuries in the elderly population towards the incidence of major intracranial injuries[3,4,5]. Statistical data from the Centres for Disease Control and Prevention (CDC) demonstrated a 17% increase in accidental falls-related TBI between 2008 and 2017 in the USA [6]. Importantly, mortality rates secondary to falls-related TBI were disproportionately higher in adults aged 75 years and above, at 54.08 per 100,000 person-year, compared with 8 per 100,000 in those aged less than 75 years. With estimates of approximately 1.5 billion individuals aged above 65 years by 2050, there is an urgent need for effective and pragmatic management of elderly patients with TBI[7]. Acute subdural haematoma (ASDH) is a potentially life-threatening form of TBI encountered by emergency services, which is characterised by extra-axial collection of acute blood between the dura and underlying brain parenchyma. The elderly are particularly vulnerable due to the risks of falls-related TBI, age-related atrophic changes in the brain, and the use of anticoagulants[8]. Hence, rationalisation of ASDH management in this group of patients is becoming increasingly pertinent[9,10]. This was highlighted by a recent single centre retrospective study in the UK, which demonstrated that the odds of inpatient death were approximately 15 times higher than its chronic counterpart in patients aged 75 years and above [11].

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In the acute setting, ASDH management consists of either urgent surgical intervention or conservative management. The former typically consists of a craniotomy and evacuation of ASDH, though a decompressive craniectomy (DC) may be required in rare instances where the brain is severely swollen intra-operatively. However, the decision for surgical intervention for ASDH in elderly patients remains a point of contention due to its associated mortality[12]. Surgical intervention is typically guided by imaging parameters including an ASDH > 10 mm thickness or midline shift (MLS) > 5 mm on computed tomography (CT) imaging, neurological deterioration, and/or intracranial pressure (ICP) monitoring[13]. Whilst these guidelines are applicable to the general management of adult TBI, their applicability to the elderly population is less clear. Quantifying surgical morbidity and mortality is vital for tailoring decisions to this age group and managing the expectations of relatives who may not appreciate the associated risks. Therefore, in this study we perform a systematic review and meta-analysis of the literature to quantify the mortality associated with surgical management of ASDH in the elderly, and identify potential risk factors for poor outcome in this group.

## Methods

This study was performed in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 Statement [14]. The protocol for this systematic review is registered on PROSPERO (CRD42020189508).

### Literature Search

A multi-database (PubMed, Embase, and Cochrane Reviews) literature search was performed between January 1990 and May 2020 by authors OM and OE (**Figure 1**). Conflict of opinion was settled by senior author MZ. The following search terms were used in varying combinations: “subdural h(a)ematoma”, “subdural h(a)emorrhage”, “outcome*”, “compar*”, “morbid*”, “mortality*”, “complication*”, “reoccur*”, “recur*”, “surg*”, “operati*”, “non(-)surgical”, “burrhole”, “crani*”, “old*”, “frail*”, “geri*”, and “elder*” (**Supplementary Data 1**). Eligible articles were restricted to English language only, and the bibliographies of included studies were screened for further relevant studies.

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**Study Selection**

Studies fulfilling the following criteria were included: (i) diagnosis of traumatic ASDH (confirmed on CT imaging), (ii) patient age of 60 years or above, (iii) patients that underwent surgical evacuation of ASDH, (iv) reported clinical outcome. Studies including patients younger than 60 years were only included if management and clinical outcomes were specifically reported for the target age group. Exclusion criteria included: (i) diagnosis of chronic subdural haematoma with no evidence of acute component; (ii) significant life-threatening extracranial injuries; (iii) abstracts, conference presentations, editorials, and expert opinions.

**Data Extraction**

The following variables of interest were extracted by authors SM and RS from eligible studies: number of patients, age, gender, baseline functional status, pre-admission residence, comorbidities, use of antithrombotic agents (antiplatelets or anticoagulants), mechanism of injury, pre-operative Glasgow Coma Score (GCS)/ pupillary reaction to light, imaging findings, indications for surgical intervention, timing of surgery, intervention performed (craniotomy or decompressive craniectomy), mortality (at discharge and longer term follow up), clinical outcome (at discharge and longer term follow up), and duration of follow up. Variables were summarised with respect to whether included studies examined associations with outcome. Studies reporting mortality and/or Glasgow Outcome scale (GOS) were considered for inclusion in meta-analysis (see below). When reported, GOS was used to dichotomise outcomes into good (GOS 4 – 5) and poor (GOS 1 – 3) categories. Studies reporting conservative management along with surgical management were examined for potential comparison between different treatment groups (see below). Studies reporting minimally invasive techniques for ASDH evacuation were included for narrative synthesis. All included studies were assessed for quality using the STROBE (Strengthening of Reporting of Observational Studies in Epidemiology) criteria[15] and an adaptation of the ROBINS-I tool [16] by two authors independently (IM and JM) (**Table 1**). Studies with STROBE score <13 or ROBINS-I score of critical risk of bias were deemed ineligible for inclusion in meta-analysis.

**Statistical Analysis**

Statistical analysis was performed using R Version 3.6.0, and the *metafor* and *meta* packages[17]. The percentage of patients under the following categories were pooled for meta-analysis: (i) deaths at discharge, (ii) deaths at longer term follow up (LTFU), (iii) poor outcome at discharge, (iv) poor outcome at LTFU. Poor outcome was defined as patients with GOS 1 - 3. Studies using other scoring systems were not included for meta-analysis to avoid introducing bias from definition of outcome. LTFU was defined as any time point beyond the time of discharge (specific duration of follow up was recorded). All studies reporting mortality and/or GOS (specific score or dichotomised) in patients aged 60 years or above that underwent surgical evacuation of ASDH were included for meta-analysis. Surgical evacuation consisted of either craniotomy or decompressive craniectomy.

Heterogeneity was quantified with the following tests: DerSimonian and Laird estimator[18] to estimate between-study variance ($\tau^2$), Cochran's Q test, and the $I^2$ statistic. Given that $I^2$ > 25% in all analyses and there was potential heterogeneity between studies with respect to specific age groups, a random effects model with double-arcsine transformation[17,19] was used. Pooled estimates were reported with 95% confidence intervals (95% CI). Studies published prior to January 2000 were excluded from meta-analysis to avoid distortion of summary effect sizes due to differences in healthcare technology as a function of time. Subgroup analysis was performed using the year of the study as a moderator (pre- and post-January 2000) with a mixed effects model[17] **(Supplementary Data 2)**. Differences between subgroup summary estimates were significantly different. Therefore, older studies were excluded on both theoretical and statistical premises. Conservative and surgical groups were not compared due to fundamental differences in treatment indications and baseline cohort characteristics, which were acknowledged by relevant studies (**Supplementary Data 3,4**).

## Results

A total of 2572 articles were screened, yielding 21 studies for final inclusion [20-40](**Table 1**). A set of fifteen studies were suitable for meta-analysis, with varying subsets used for each separate analysis depending on available outcome data. All studies were single centre retrospective studies, except for a recent retrospective multicentre study performed in Italy involving 213 patients from 5 centers[38].

### Mortality

Mortality was generally defined as death by a specific time point, except for one study that also included death or discharge to a hospice[34]. Mortality at discharge (381/820 patients) and LTFU (322/622 patients) was reported by 13 studies, of which 10 studies were included for meta-analysis (**Figure 2**). Duration of follow up was provided by all studies that reported mortality at LTFU, with an mean of 7.4 months (range 2 - 18 months). The pooled estimate of percentage of deaths was 39.83% (95% CI: 32.73, 47.14%) at discharge, and 49.30% (95% CI: 42.01, 56.61%) at LTFU. Mean duration of follow up in studies included for meta-analysis was 7.1 months (range 2 - 12 months) (**Table 2**).

### Functional Outcomes

Poor outcome was defined as patients with GOS 1 - 3. GOS on discharge (poor outcome in 437/532 patients) was reported by 9 studies, of which 6 were included for meta-analysis. GOS at LTFU (poor outcome in 459/578 patients) was reported by 11 studies, of which 8 were included for meta-analysis (**Figure 2**). Duration of follow up was provided by all studies that reported GOS at LTFU, with a mean of 6.8 months (range 2 - 18 months). Pooled estimate of percentage of poor outcomes was 81.18% (95% CI: 75.61, 86.21%) at discharge, and 79.25% (95% CI: 72.42, 85.37%) at LTFU. Mean duration of follow up in studies included for meta-analysis was 6.4 months (range 2 - 12 months) (**Table 2**). One study[37] reported modified Rankin Scale (mRS) scores at discharge and LTFU. Poor outcome was defined as mRS 5 - 6, and reported in 28% (8/28 patients) at discharge and 57% (16/28 patients) at LTFU.

## Risk factors for poor outcome

*Demographics*

Inclusion age was reported by all studies, and analysed with respect to outcome in older age groups in 9 studies (**Table 3**). Of these, 2 studies reported a significant effect of age on outcome[31,34], whilst the remainder reported no significant associations. With respect to GOS at 30 days, age was negatively associated with GOS on multivariate (RC -0.054, p = 0.013; other factors included volume of ASDH, GCS, MLS, and antiplatelet use) but not univariate linear regression (RC - 0.043, p = 0.095) analyses[31]. Age was negatively associated with GOS at 6 months on both univariate (RC -0.063, p = 0.031) and multivariate (RC -0.077, p = 0.002) linear regression analyses[31]. Another study[34] demonstrated a significant association between patient age group (70-79 years, 80 years and above) and mortality on univariate analysis (p = 0.05; Chi-squared test) and age (70-79 years vs 80 years and above) as a predictor of mortality on multivariate logistic regression (OR 2.83; 95% CI: 1.18, 6.83; p = 0.02; other covariates included GCS and surgical approach). Gender was reported by 15 studies, and analysed with respect to outcome in 4 studies (**Table 3**). A significant association with outcome was not demonstrated in any of these studies[30,36,38,39].

*Baseline functional status*

Baseline functional status was reported in 11 studies, of which 5 studies examined association with outcomes (**Table 3**). One study[39] included patients aged 80 years and above demonstrated a significant association between outcome (favourable, unfavourable) and number of co-morbidities (more than 1, less than or equal to 1) (p = 0.05; Fisher's exact test). This was not statistically significant on analysis with a greater number of co-morbidities (more than or equal to 5, less than 5). In contrast, other studies reported no significant association between number of co-morbidities and dichotomised GOS[32,38] or mortality[34]. Two studies demonstrated no statistically significant differences between survivors and non-survivors with respect to pre-morbid independence[30,34] or pre-admission residence[30].

*Use of anti-thrombotic agents*

Use of anti-thrombotic agents was reported by 13 studies, of which 9 studies examined statistical association with outcome (**Table 3**). Trevisi et al.[38] demonstrated a significant

association between use of anti-thrombotic agents and dichotomised GOS on univariate analysis (82 vs 57%; good vs poor outcome; p = 0.01), though multivariate logistic regression demonstrated no significant effect of anti-thrombotic use on dichotomised GOS (OR 0.33; 95% CI: 0.10, 1.10; p = 0.02; other covariates included GCS, ASDH thickness, and MLS). Other studies demonstrated no significant association between anti-thrombotic use and GOS[26,35], dichotomised GOS[32,36], or mortality[30,35]. Examination of antiplatelet and anticoagulant use in isolation also demonstrated no significant association with GOS[31,39] or mortality[30,34].

*Mechanism and Severity of Injury*

Mechanism of injury was reported in 7 studies [22,26-28,30,34,35]. The association between mechanism of injury and outcome was examined in 5 of these studies, but a significant association was not demonstrated [22,26,28,30,34]. In all 7 studies reporting mechanism of injury, ground-level falls were the most common mechanism (range 65 – 100%), with road traffic collisions being the second most common albeit much less common than falls (range 4 – 30%).

Neurological status prior to surgical intervention was reported in all 21 included studies. Association between GCS or pupillary reactivity and outcome were examined in 11 studies. Of these, 10 studies demonstrated significant associations (**Table 3**). Neurological status was reported pre-operatively in 2 studies[22,35], on admission and pre-operatively in 2 studies[36,38], and on admission in the remaining 6 studies. Significant heterogeneity was observed in the methodology used to assess the relationship between pre-operative neurological status and outcome (**Supplementary Data 5, 6**). Statistically significant associations between GCS and GOS[31,38,39] or mortality[26,30,33-35] were demonstrated across multiple studies. Similarly, significant associations were also demonstrated between pupillary reactivity and GOS[22,26,35,36] or mortality[26,35]. No significant relationship between GCS and GOS was demonstrated in four studies[22,32,33,36], and between pupil reactivity and GOS/ mortality in four studies[30,32,38,39].

Extracranial injuries were reported in 4 studies[21-23,29], while patients with significant extracranial injuries were excluded from five studies[20,24,25,31,37]. There were no reports of

extracranial injuries in the remaining studies. None of the included studies examined the relationship between extracranial injuries and outcome. However, early studies demonstrated poor outcomes in several patients with concomitant injuries: (i) of six patients with orthopaedic fractures (three patients with femoral fractures, two patients with rib fractures and one patient with a C2 fracture), five patients died[21]; and (ii) all eight patients with orthpaedic fractures had poor outcomes[22].

*Imaging*

Imaging findings with respect to ASDH were reported in 15 included studies. The association between imaging findings and outcome was examined in eight of these studies, of which five showed significant findings (**Table 3**). MLS was reported in 7 studies[26,31,32,35,36,38,39] , while ASDH thickness or volume (estimated by volumetric analysis of CT scans) was reported in 6 studies[26,31,32,36,38,39]. All studies reporting these imaging parameters analysed their association with outcome.

Petridis et al.[26] demonstrated a significantly greater proportion of patients with MLS >10mm in the GOS-1 than in the GOS-5 category (Chi-square test; p<0.005). Furthermore, the difference between ASDH thickness and MLS was quantified and its association with outcome was examined. A significant association was identified between mean ASDH-MLS difference and individual GOS groups (ANOVA; p <0.005), measuring 6.3 mm in the GOS-5 and 1.8 mm in the GOS-1 category. Similarly, three other studies demonstrated a significant association between mean MLS and dichotomised GOS[31,38] or mortality[35]. One study demonstrated a negative association between MLS and GOS at both 30 days (RC -0.053, p = 0.015) and 6 months (RC -0.055, p = 0.026) on univariate linear regression, though this did not reach statistical significance on multivariate analysis[31]. Two studies demonstrated no significant association between MLS and dichotomised[39] or trichotomized[35] GOS.

Two studies demonstrated significant associations between ASDH volume and outcome[31,39]. One study demonstrated a negative association between ASDH volume and GOS at both 30 days (RC -0.006, p = 0.005) and 6 months (RC -0.005, p = 0.027) on univariate linear regression, though this did not reach statistical significance on multivariate analysis[31].

Another study demonstrated a significantly smaller ASDH volume in the favourable outcome group when compared with the unfavourable outcome group (67ml vs 118ml; unpaired Student's *t* test, p=0.05) [39]. Two studies demonstrated no significant association between ASDH thickness and GOS categories[26] or dichotomised GOS[38].

The presence of additional intracranial injuries, such as cerebral contusions and traumatic subarachnoid haemorrhage, were reported in 8 studies[21-24,26,35,37-39] but only statistically analysed with respect to outcome in 3 studies[22,26,35]. None of these studies demonstrated significant associations between additional intracranial injuries and outcome.

*Surgical factors*

Timing of surgery from the time of admission or injury was reported in 10 studies, and analysed with respect to outcome in 4 studies (**Table 3**). No statistically significant associations were demonstrated with respect to dichotomised GOS in 3 studies[22,36,39]. One study demonstrated a significant association of timing of surgery after admission, divided into 6 hour increments between <6 hours and >72 hours, and GOS (Chi-squared test; $p < 0.001$)[38]. However, on univariate logistic regression, timing of surgery (within 6 hours or >6 hours), demonstrated no significant effect on prediction of dichotomised GOS ($p = 0.20$).

Choice of surgical approach included craniotomy or DC. Four studies did not distinguish between craniotomy and decompressive craniectomy, but defined surgical intervention as the surgical evacuation of ASDH [20-23]. Of the remaining studies, 12 studies included craniotomy and DC, 4 studies included craniotomies alone, and 1 study defined surgical intervention as craniectomies [24] (**Table 1**). Number of patients undergoing each surgical approach was specifically reported in 6 studies, however, only 3 studies examined association with outcome (**Table 3**). In total, 52 patients underwent DC, but outcomes were only reported for 23 patients [32,34,36]. Two studies reported poor outcomes in in 2 out of 2, and 8 out of 10 patients [32,36]. One study reported mortality in 9 out of 11 patients [34]. Monsivais et al.[34] demonstrated a significant association between surgical approach (craniotomy, DC) and mortality on univariate analysis (Fisher's exact test; $p<0.01$), and DC as a predictor of mortality when compared with craniotomy on multivariate logistic regression

(OR 5.72; 95% CI: 1.11, 29.32; p = 0.04; other covariates included age and GCS). Other studies demonstrated no significant associations between choice of surgical approach and dichotomised GOS[32,36] or mortality[36].

## Discussion

*Surgical Outcomes*

ASDH is a devastating traumatic pathology, and particularly so in the elderly. Although the significant mortality and morbidity associated with surgical intervention has been recognised for decades, there are no clear guidelines for ASDH management in this age group. With a growing elderly population worldwide and a shift in healthcare policy towards prioritising quality of life, it is essential that neurosurgical guidelines are adapted accordingly[9,10]. Indeed, the generic guidance for surgical intervention based on ASDH thickness, MLS, and neurological deterioration may not necessarily be appropriate for elderly patients with ASDH[13]. In this study, we systematically reviewed the literature for studies reporting outcomes following surgical intervention in patients aged 60 years and above with ASDH. We demonstrate that the pooled estimated mortality rate in this group is 40% at the point of discharge, and 49% at LTFU (**Figure 2**). Furthermore, the pooled estimated poor outcome (GOS 1 – 3) rate is 81% at discharge and 79% at LTFU. Analysis of separate age subgroups could not be performed due to insufficient reported data, and therefore inclusion age of above 60 years was used as a minimum criterion for consideration. Only one study included patients above 60 years of age[25], which was excluded from meta-analysis due to its date of publication. Therefore, our pooled estimated mortality and unfavourable outcome rates are generated from studies including patients aged 65 years and above. One potential source of heterogeneity was the inclusion of both craniotomy and DC within meta-analysis, as the majority of included studies did not distinguish reliably between surgical approach. Despite 11 studies including craniotomy and DC, only 7 studies reported specific numbers for each approach, and outcome could not be consistently deduced. We decided to include both procedures within the meta-analysis because: (i) we aimed to provide a pooled estimate for surgical intervention for ASDH in the elderly; and (ii) the decision between craniotomy and DC is intra-operative, therefore

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quantification of mortality and morbidity following either procedure remains pertinent. In addition, of the three studies examining effect of surgical approach on GOS or mortality, only one study demonstrated a significant association with mortality[34]. However, this was likely due to the small proportion of patients that underwent DC. Another source of heterogeneity was the duration of follow up. For the pooled estimate of mortality at LTFU, duration of follow up was 6 months in five studies, 12 months in three studies, 3 months in one study, and 2 months in one study. Similarly, for the pooled estimate of poor GOS at LTFU, duration of follow up was 6 months in four studies, 12 months in two studies, 3 months in one study, and 2 months in one study. Therefore, pooled mortality may have been underestimated given that a higher percentage of deaths would be expected at longer durations of follow up. Whilst the same concept may be true for pooled estimated poor outcomes, it is also possible that shorter durations of follow up did not capture potential future neurological recovery. Given that included studies did not consistently report the direct cause of poor outcome or mortality, it remains difficult to extrapolate any meaningful conclusions regarding this factor. Surgical inclusion or exclusion criteria were detailed in ten studies, and largely revolved around existing Brain Trauma Foundation Guidelines[13] (**Supplementary Data 4**). The possibility of comparing surgical and conservative was initially evaluated, as four studies reported outcomes in conservatively managed patients [28,33,37,38]. However, indications for conservative management were due to inoperable co-morbidities, good neurological status, moribund clinical status, or unspecified (**Supplementary Data 3**). Therefore, surgical and conservative groups were not feasible for comparison.

*Risk Factors*

We also examined risk factors for mortality and poor outcome in the target cohort. Of 9 studies examining the relationship between age and outcome, only 2 studies demonstrated a statistically significant effect[31,34]. One study demonstrated a negative association between age and GOS at 30 days and 6 months on multivariate regression[31]. Whether GOS should be modelled as a continuous variable is arguable, however, as each increment does not represent an equal difference in functional status. Another study, however, demonstrated

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that the odds of death were almost three times higher in patients aged 80 years and above compared with patients aged 70 – 79 years. Whilst age appears intuitively likely to be associated with poorer outcomes, few studies demonstrate a significant association in the elderly cohort. Only two studies examined the relationship between baseline functional status and outcome. Neither demonstrated any difference between survivors and non-survivors with respect to pre-morbid independence[30,34] or pre-admission residence[30]. Further studies are required to elucidate this relationship, since the use of scoring systems such as Karnofsky Performance Status[41] or Clinical Frailty Score[42] may provide a feasible method to prognosticate surgical candidates. This has been demonstrated by a recent study, which highlighted the use of the Modified Frailty Index[43] to predict 30 day mortality and 6 month unfavourable outcome[44]. Similarly, four studies reported the relationship between co-morbidities and outcome, though only one study used a formal scoring system for co-morbidities[38]. However, no significant association with dichotomised GOS was demonstrated[38]. In a similar fashion to baseline functional status, further studies using scoring systems such as the Charlston Co-morbidity Index[45] would be useful for assessment for neurosurgical decision-making.

In contrast, the association between neurological status and outcome were examined in 11 studies. Although significant associations were demonstrated across several studies, a considerable degree of heterogeneity was observed in the approach to analysis (**Supplementary Data 5, 6**). GCS was modelled as a continuous variable, dichotomised at various points, or categorised with respect to severity of TBI. However, the emerging theme was that a lower GCS on admission was associated with a greater risk of mortality. Some studies demonstrated a similar relationship with risk of poor GOS[31,38,39], though this did not consistently reach statistically significance[22,32,33,36]. Therefore, identifying a cut-off for admission GCS could be an important avenue for future research to help identify patients that are highly unlikely to benefit from surgery. Pupillary reactivity to light was also examined across several studies. As expected, fixed and dilated pupils were often associated with poor outcomes and mortality (**Supplementary Data 6**), although some studies revealed

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no significant association. This could be due to subjective errors on reporting of pupillary reactivity, which may be remedied in future studies with use of pupillometers. In the absence of a reliable cut-off for pre-operative GCS, understanding the relationship between pupillary reactivity and outcome is essential for prognosticating intubated patients. Several studies examined the relationship between imaging findings and outcome, demonstrating a significant association between MLS and outcome in four studies[26,31,35,38]. Similar to the assessment of pre-operative neurology, approaches to analysis were varied, with no reliable conclusion with respect to a prognostic cut-off value for MLS. Interestingly, whilst ASDH thickness was not significantly associated with outcome [26,38], ASDH volume was significantly associated with GOS in two studies[31,39]. The reported negative association between ASDH-MLS differential and GOS may also warrant further investigation[26]. This has been previously reported in a retrospective cohort study of ASDH patients, which demonstrated a 75% mortality rate in patients with an ASDH-MLS differential of >5mm[46]. In contrast, the group with an ASDH-MLS differential of 0mm only had a 25% mortality rate. Therefore, further studies comparing different measurement approaches may yield useful information regarding prognostication.

*Limitations*

We acknowledge several limitations from our systematic review and meta-analysis. Although we performed a multi-database literature search, ten full text articles could not be accessed, which may have reported valid data for inclusion. Furthermore, non-English studies were excluded, raising the potential for geographical bias. Despite assessment of risk factors across included studies, a subgroup analysis could not be performed due to significant variation in the reporting of relevant data. Future studies should ensure that this is addressed as developing risk stratification scores will be essential to identify suitable surgical candidates. Also, duration of follow up varied between studies with a range of 2 – 18 months, therefore, pooled estimates of outcomes at LTFU do not give an accurate indication of timescale. The majority of included studies were deemed at moderate risk of bias, though two studies were at serious risk (**Supplementary Data 7**). However, none of the included studies were at critical risk of bias, and were therefore eligible for inclusion in

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meta-analysis. Of note, two key studies[47,48] could not be included due to failure to meet our eligibility criteria. Shibahashi et al.[47] compared outcomes following craniotomy and DC for ASDH across 1028 patients matched for age, gender, admission GCS and vital signs, and comorbidities. Interestingly, no differences in mortality between treatment groups in the elderly cohort were found. Indeed, this finding supports our approach to a pooled estimate for mortality on meta-analysis. This study was not included in our review due to it not presenting absolute mortality figures, rather a mean difference between groups only. While it provides some evidence of no difference in mortality when comparing craniotomy with DC, the groups were not matched for pupillary abnormalities, nor were any imaging parameters taken into account in the matching process. One recent study was published outside of the dates of our search strategy, examining a total of 2508 patients aged 65 years and older that underwent surgical evacuation of ASDH [48]. Data was collected from the USA National Trauma Data Bank registry between 2016 and 2017, which spans approximately 765 centres. Interestingly, inpatient mortality was estimated at only 30.5%. Multivariate logistic regression revealed that age, pre-operative GCS, traumatic subarachnoid haemorrhage, and MLS > 5 mm were significant predictors of mortality. Although the risk factors are in keeping with findings from previous studies, the reported inpatient mortality was comparatively low. Given that surgical approach was not specified, as ICD-10 diagnosis codes were used to extract data, and surgical indications may have varied significantly between centres, it is unclear whether inpatient mortality is truly representative of this population.

## Conclusion

In this systematic review and meta-analysis, we demonstrate that outcomes following surgical evacuation of ASDH in patients aged 65 years and above are poor. Pooled estimated mortality rates in this group are 40% at discharge and 49% at LTFU. Estimated rates of poor GOS are 81% at discharge and 79% at LTFU. Potential risk factors for poor outcome include age, baseline functional status, pre-operative neurological status, and imaging parameters. Whilst further studies are required to reliably characterise predictors of poor outcome to inform separate guidelines for surgical intervention in this cohort, this systematic review constitutes the best level of evidence in the current literature that surgical intervention for

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ASDH in the elderly carries significant risks, which must be weighed against benefits. We, herein, provide robust evidence that helps healthcare professional decision-making and counselling of families on a contentious issue.

**Contributions**

**SM**- data curation, formal analysis, project administration, software, writing- original draft; **RS**- data curation, formal analysis, project administration, software, writing- original draft; **OM**- investigation, methodology; **IM**- investigation, methodology; **OE**- investigation, methodology; **JM**- investigation, methodology, writing- reviewing & editing; **MZ**- conceptualization, supervision, writing- review & editing

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**Competing Interests Statement**

Authors have no competing interests to declare.

**Data Availability**

Available on request to corresponding author.

**Patient and Public involvement**

There was no patient or public involvement in this research study.

**Ethics**

No ethical approval was required for this study as it does not involve identifiable human data use or the use of animal subjects.

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**Table 1.** Summary of included studies of ASDH in elderly following systematic review of the literature. Abbreviations: DC- decompressive craniectomy, GCS- Glasgow Coma Score, GOS- Glasgow Outcome Scale, ICP- intracranial pressure, SDH- subdural haematoma.

| Author, Year | Location (Period) | Country | No. of hospitals/ Patients | Age for inclusion (years) | Age (years) % Male | Treatment | Outcomes measured | Risk factor analysis | STROBE criteria | ROBINS-I score |
|---|---|---|---|---|---|---|---|---|---|---|
| Wilberger et al., 1991[20] | Allegheny General Hospital (1982 to 1987) | USA | 1/ 28 | > 65 | N/A | Craniotomy | Mortality, GOS at discharge | Age, ICP, GCS, timing of surgery; descriptive analyses (not in elderly group alone) | 17 | Moderate |
| Cagetti et al., 1992[21] | University of Genoa Medical School (Jan 1980 to Dec 1988) | Italy | 1/ 26 | > 80 | 84.2 (mean) 57.7% | Surgical evacuation | Mortality, GOS at discharge | Age, mechanism, GCS, comorbidities; descriptive analyses (not in elderly group alone) | 10 | Serious |
| Jamjoom, 1992[22] | Frenchay Hospital (1980 to 1989) | UK | 1/ 27 | >= 75 | 79.2 (mean) 55.6% | Surgical evacuation | Mortality, GOS at 6 months | Age, sex, mechanism, GCS/ pupils, imaging, timing of surgery; Chi-squared test | 14 | Moderate |
| Kotwica & Jakubowski, 1992[23] | Medical University of Lodz (1984 to 1990) | Poland | 1/ 27 | > 70 | N/A | Surgical evacuation | Mortality, GOS at discharge | Age, gender, GCS, imaging; descriptive analyses | 15 | Moderate |

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|---|---|---|---|---|---|---|---|---|---|---|
| Massaro et al., 1996[24] | Centro Traumatologico Ortopedico (1982 to 1992) | Italy | 1/ 25 | > 65 | N/A | DC | Mortality, GOS at 18 months | Age, gender, mechanism, GCS, timing of surgery, imaging; Chi-squared test/ Fisher's exact test (not in elderly group alone) | 15 | Moderate |
| Koc et al., 1997[25] | Erciyes University (Jan 1986 to Aug 1995) | Turkey | 1/ 15 | >= 61 | N/A | Craniotomy | Mortality, GOS at 3 months | Age, gender, presenting GCS; Chi-squared test (not in elderly group alone) | 10 | Moderate |
| Petridis et al., 2009[26] | UKSH Campus Kiel | Germany | 1/ 119 | > 65 | N/A<br>55.2% | Craniotomy, DC | Mortality, GOS discharge | GCS/ pupils, imaging, anticoagulation, ICP; Chi-squared test, ANOVA | 13 | Moderate |
| Hanif et al., 2009[27] | Beaumont Hospital (Jan 1999 to Dec 2003) | Ireland | 1/ 29 | > 70 | 77.0 (mean)<br>59.7% | Surgical evacuation | Mortality, GOS at 6 months | Age, GCS; descriptive analyses (not in elderly group alone) | 13 | Moderate |
| Taussky et al., 2012[28] | Kontansspital Aarau (Jan 2002 to Dec 2007) | Switzerland | 1/ 37 | > 65 | 73.0 (median)<br>46% | Craniotomy - 23<br>DC - 14 | Mortality at discharge<br>Mortality, GOS at 6 months | Age, pupils, timing of surgery, GCS/ pupils, anticoagulants, co-morbidities; descriptive analyses | 20 | Moderate |

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|---|---|---|---|---|---|---|---|---|---|---|
| Hamed et al., 2016[40] | Rheinische Friedrich-Wilhelms University (Jan 2010 to Dec 2014) | Germany | 1/57 | >70 | N/A | Craniotomy, DC | GOS at 6 months | Age, TBI severity, timing of admission, surgical approach, anti-thrombotics, timing of surgery, duration of hospital stay, post-operative complications (not for elderly group alone) | 17 | Serious |
| Merzo et al., 2016[29] | Uppsala University Hospital (2008 to 2010) | Sweden | 1/ 24 | >= 65 | N/A | Craniotomy – 23 DC – 1 | Mortality, GOS at 6 months | Age, gender, comorbidities, mechanism, imaging; Chi-squared, unpaired Student's t-test (not for elderly group alone) | 17 | Moderate |
| Raj et al., 2016[30] | Helsinki University Hospital (Jan 2009 to Dec 2012) | Finland | 1/ 44 | >= 75 | 81.0 (median) 52% | Craniotomy, DC | Mortality at 1 and 3 years | Age, baseline functional status, residence, gender, anticoagulants, mechanism, GCS/ pupils; Chi-squared test, unpaired Student's t test, Mann Whitney U test | 18 | Moderate |
| Benedetto et al., 2017[31] | Azienda Ospedaliero-Universitaria Pisana (June 2011 to Dec 2014) | Italy | 1/ 67 | > 70 | 80.5 (median) 53.7% | Craniotomy | Mortality, GOS at 1 and 6 months | Age, imaging, GCS, antiplatelets; simple and multiple linear regression | 17 | Moderate |

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|---|---|---|---|---|---|---|---|---|---|---|
| McGinity et al., 2017[32] | University of Texas Health Science Center (2005 to 2015) | USA | 1/ 34 | >= 80 | 84.0 (mean) N/A | Craniotomy – 32 DC – 2 | Mortality, GOS at 2 months (variable) | Age, mechanism, type of surgery, anticoagulants, GCS/ pupils, imaging, comorbidities; Fisher's exact test, Mann Whitney U test | 17 | Moderate |
| Won et al., 2017[33] | Goethe-University Hospital (Jan 2007 to Dec 2016) | Germany | 1/ 56 | >= 80 | 85.0 (mean) 43% | Craniotomy, DC | Mortality at discharge | GCS; univariate/ multivariate logistic regression | 20 | Moderate |
| Monsivais et al., 2018[34] | University of Texas Health Science Centre (Jan 2006 to Jul 2016) | USA | 1/ 112 | >= 70 | N/A | Craniotomy – 101 DC – 11 | Mortality at discharge | Age, GCS, type of surgery; Chi-squared test/ Fisher's exact test | 20 | Moderate |
| Akbik et al., 2019[35] | University of New Mexico Hospital (Jan 2013 to Dec 2017) | USA | 1/ 62 | >= 65 | 78.0 (median) 48% | Craniotomy, DC | Mortality, GOS at discharge and 3 months | Age, comorbidities, GCS/ pupils, imaging, antithrombotics; Kruskal Wallis, Fisher's exact test | 15 | Moderate |

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|---|---|---|---|---|---|---|---|---|---|---|
| Bus et al., 2019[36] | Academic Medical Centre (Jan 2000 to Oct 2015) | Netherlands | 1/ 84 | >= 65 | 75.0 (mean)<br>64.3% | Craniotomy – 74<br>DC – 10 | Mortality at discharge Mortality, GOS at 1 year | Age, gender, imaging, GCS/ pupils, age of SDH, imaging, timing of surgery, anticoagulation; Chi-squared/ Fisher's exact tests, Mann Whitney U test, univariate logistic regression | 17 | Moderate |
| Sufaro et al., 2019[37] | Soroka University Medical Center (2006 to 2016) | Israel | 1/ 28 | > 70 | 81.5 (mean)<br>53.6% | Craniotomy | Mortality, mRS score at discharge and 1 year | Age, gender, baseline status, antithrombotics, GCS, imaging, neurological deficit; descriptive analyses | 17 | Serious |
| Trevisi et al., 2020[38] | Ospedale Santo Spirito; Fondazione Policlinico Universitario A. Gemelli IRCSS; S. Anna University Hospital, UOC Neurochirurgia (Jan 2016 to Dec 2019) | Italy | 5/ 147 | >= 70 | 78 (mean)<br>58% | Craniotomy-133<br>DC- 14 | Mortality, GOS at discharge and 6 months | Age, gender, comorbidities, GCS/ pupils, imaging, side of SDH, size of craniotomy, timing of surgery, anticoagulants; Chi-squared test, univariate/ multivariate logistic regression | 16 | Moderate |

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| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Younsi et al., 2020[39] | University Hospital Heidelburg (Jan 2006 to Dec 2016) | Germany | 1/ 27 | >= 80 | 84.0 (median) 60% | Craniotomy, DC | Mortality, GOS at discharge | Age, gender, co-morbidities, GCS/ pupils, imaging, surgical factors, anticoagulants; Fisher's exact test, unpaired Student t test | 21 | Serious |

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**Table 2.** Mortality, outcomes, and duration of follow up following surgical evacuation of ASDH in patients aged 60 years and above.

Abbreviations: GOS- Glasgow Outcome Scale.

| Author, Year | No. of Patients | *At Discharge* (N, %) | | | *At Long Term Follow Up* (N, %) | | | |
|---|---|---|---|---|---|---|---|---|
| | | Deaths | GOS 1 – 3 | GOS 4 – 5 | Deaths | GOS 1 – 3 | GOS 4 – 5 | Duration |
| Wilberger et al., 1991[20] | 28 | 23 | - | - | - | - | - | - |
| Cagetti et al., 1992[21] | 26 | 23 | 23 | 3 | - | - | - | - |
| Kotwica & Jakubowski, 1992[23] | 27 | 23 | - | - | - | - | - | - |
| Jamjoom, 1992[22] | 27 | - | - | - | 19 | 23 | 4 | 6 |
| Massaro et al., 1996[24] | 25 | - | - | - | 15 | 22 | 3 | 18 |
| Koc et al., 1997[25] | 15 | - | - | - | 11 | 12 | 3 | 3 |
| Petridis et al., 2009[26] | 119 | 64 | 94 | 25 | - | - | - | - |
| Taussky et al., 2012[28] | 37 | 13 | - | - | 13 | 22 | 15 | 6 |
| Raj et al., 2015[30] | 44 | - | - | - | 22 | - | - | 12 |
| Merzo et al., 2016[29] | 24 | - | - | - | 9 | 19 | 5 | 6 |
| Hamed et al., 2016[40] | 57 | - | - | - | - | 53 | 4 | 6 |
| Benedetto et al., 2017[31] | 67 | 37 | 60 | 7 | 45 | 58 | 9 | 6 |
| Won et al., 2017[33] | 56 | 13 | - | - | - | - | - | - |

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| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| McGinity et al., 2017[32] | 33 | - | - | - | 18 | 27 | 6 | 2 |
| Monsivais et al., 2018[34] | 112 | 47 | - | - | - | - | - | - |
| Akbik et al., 2019[35] | 62 | 24 | 54 | 8 | 27 | 45 | 17 | 3 |
| Bus et al., 2019[36] | 84 | 44 | - | - | 54 | 63 | 21 | 12 |
| Sufaro et al., 2019[37] | 28 | 6 | - | - | 12 | - | - | 12 |
| Trevisi et al., 2020[38] | 147 | 51 | 114 | 33 | 64 | 115 | 32 | 6 |
| Younsi et al., 2020[39] | 27 | 9 | 21 | 6 | - | - | - | - |

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**Table 3.** Potential risk factors for mortality or poor outcome following surgical evacuation of ASDH in patients aged 60 years and above. N- not reported; X- reported but not statistically analysed; NS- non-significant association on statistical analysis; S- significant association on statistical analysis. Abbreviations: DC- decompressive craniectomy, GCS- Glasgow Coma Score, GOS- Glasgow Outcome Scale.

| Author, Year | Outcome | Intervention | Risk Factors | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Age | Sex | Base-line | GCS/ pupils | Imaging | Mechanism of Injury | Other intracranial injuries | Timing of surgery | Craniotomy vs DC | Antithrombotic agents |
| Wilberger et al., 1991[20] | Mortality GOS | Surgical | X | X | N | X | N | N | N | X | N | N |
| Cagetti et al., 1992[21] | Mortality GOS | Surgical | X | X | X | X | N | N | X | N | N | N |
| Jamjoom, 1992[22] | GOS | Surgical | NS | X | N | S | NS | NS | NS | NS | N | N |
| Kotwica & Jakubowski, 1992[23] | GOS | Surgical | X | X | N | X | X | N | X | N | N | N |
| Massaro et al., 1996[24] | Mortality GOS | DC | X | X | N | X | X | N | X | X | N | N |
| Koc et al., 1997[25] | Mortality GOS | Craniotomy | X | X | N | X | X | N | N | N | N | N |
| Hanif et al., 2009[27] | Mortality GOS | Craniotomy | X | X | N | X | N | X | N | N | N | N |
| Petridis et al., 2009[26] | Mortality GOS | Craniotomy, DC | X | X | N | S | S | NS | NS | N | NS | NS |
| Taussky et al., 2012[28] | GOS | Craniotomy, DC | X | X | X | X | N | NS | N | X | X | X |
| Hamed et al., 2016[40] | GOS | Craniotomy, DC | X | N | N | X | X | N | N | X | X | X |

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| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Merzo et al., 2016[29] | Mortality GOS | Craniotomy, DC | X | N | X | X | N | N | N | N | X | N |
| Raj et al., 2016[30] | Mortality | Craniotomy, DC | NS | NS | NS | S | N | NS | N | N | N | NS |
| Benedetto et al., 2017[31] | GOS | Craniotomy | S | X | X | S | S | N | N | N | X | NS |
| McGinity et al., 2017[32] | GOS | Craniotomy, DC | NS | N | NS | NS | NS | N | N | N | NS | NS |
| Won et al., 2017[33] | Mortality GOS | Craniotomy, DC | X | X | X | S | X | N | N | X | N | X |
| Monsivais et al., 2018[34] | Mortality | Craniotomy, DC | S | N | NS | S | N | NS | N | X | S | NS |
| Akbik et al., 2019[35] | Mortality GOS | Craniotomy, DC | NS | X | N | S | S | X | NS | N | N | NS |
| Bus et al., 2019[36] | Mortality GOS | Craniotomy, DC | NS | NS | N | S | NS | N | N | NS | NS | NS |
| Sufaro et al., 2019[37] | Mortality mRS | Craniotomy | X | X | X | X | X | N | X | N | N | X |
| Trevisi et al., 2020[38] | GOS | Craniotomy, DC | NS | NS | NS | S | S | N | X | S | X | S |
| Younsi et al., 2020[39] | GOS | Craniotomy, DC | NS | NS | S | S | S | N | X | NS | N | NS |

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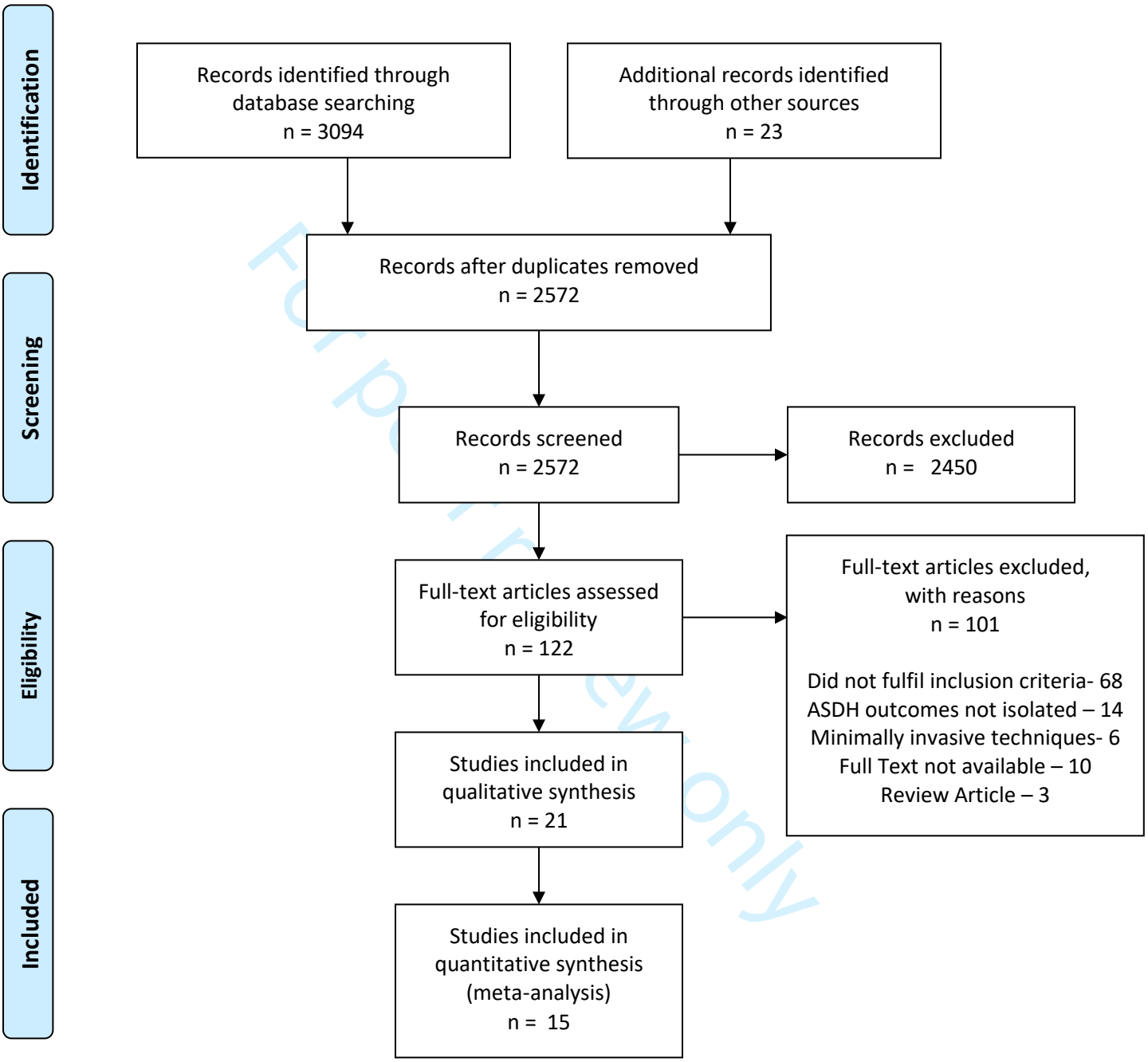
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## Figure Legends

**Figure 1.** Search strategy for systematic literature review in accordance with PRISMA guidelines.

**FIgure 2.** Forest plots depicting pooled estimates of mortality at discharge (A), poor outcome at discharge (B), mortality at discharge (C), and mortality at long term follow up (LTFU) (D) in patients aged 65 years and above that underwent surgical evacuation of ASDH.

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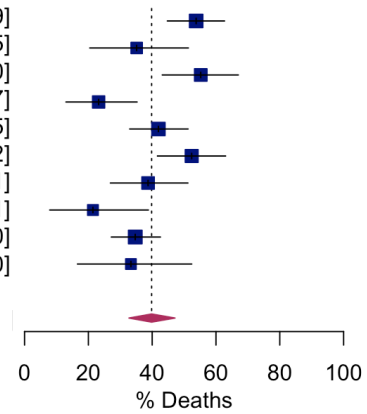

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**A**

## Deaths at discharge

| Study | Deaths | Total | % Deaths | 95% C.I. |
|---|---|---|---|---|
| Petridis 2009 | 64 | 119 | 53.78 | [44.75; 62.69] |
| Taussky 2012 | 13 | 37 | 35.14 | [20.44; 51.35] |
| Benedetto 2017 | 37 | 67 | 55.22 | [43.15; 67.00] |
| Won 2017 | 13 | 56 | 23.21 | [12.97; 35.27] |
| Monsivais 2018 | 47 | 112 | 41.96 | [32.95; 51.25] |
| Bus 2018 | 44 | 84 | 52.38 | [41.63; 63.02] |
| Akbik 2019 | 24 | 62 | 38.71 | [26.90; 51.21] |
| Sufaro 2019 | 6 | 28 | 21.43 | [ 7.89; 38.81] |
| Trevisi 2020 | 51 | 147 | 34.69 | [27.19; 42.60] |
| Younsi 2020 | 9 | 27 | 33.33 | [16.58; 52.40] |
| **Total** | **308** | **739** | | |
| **Random effects model** | | | **39.83** | **[32.73; 47.14]** |

Heterogeneity: $I^2 = 73\%$, $\tau^2 = 0.0095$, $\chi^2_9 = 33.75$ ($p < 0.01$)

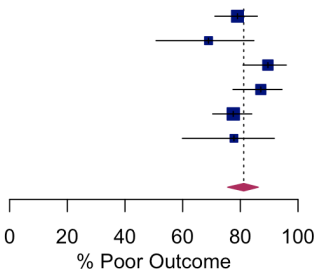



**B**

## Poor outcome at discharge

| Study | Poor Outcome | Total | % Poor Outcome | 95% C.I. |
|---|---|---|---|---|
| Petridis 2009 | 94 | 119 | 78.99 | [71.17; 85.88] |
| Hanif 2009 | 20 | 29 | 68.97 | [50.77; 84.71] |
| Benedetto 2017 | 60 | 67 | 89.55 | [80.93; 95.92] |
| Akbik 2019 | 54 | 62 | 87.10 | [77.45; 94.46] |
| Trevisi 2020 | 114 | 147 | 77.55 | [70.42; 83.96] |
| Younsi 2020 | 21 | 27 | 77.78 | [59.90; 91.79] |
| **Total** | **363** | **451** | | |
| **Random effects model** | | | **81.18** | **[75.61; 86.21]** |

Heterogeneity: $I^2 = 45\%$, $\tau^2 = 0.0029$, $\chi^2_5 = 9.12$ ($p = 0.10$)

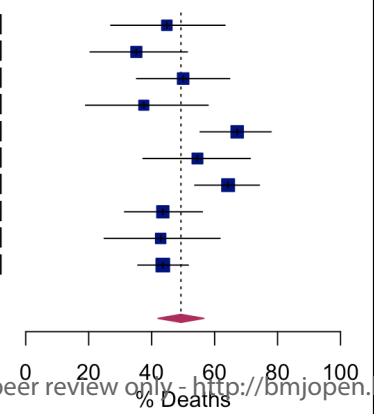



**C**

## Deaths at Long Term Follow Up

| Study | Deaths | Total | % Deaths | 95% C.I. |
|---|---|---|---|---|
| Hanif 2009 | 13 | 29 | 44.83 | [27.00; 63.32] |
| Taussky 2012 | 13 | 37 | 35.14 | [20.44; 51.35] |
| Raj 2015 | 22 | 44 | 50.00 | [35.19; 64.81] |
| Merzo 2016 | 9 | 24 | 37.50 | [19.00; 57.95] |
| Benedetto 2017 | 45 | 67 | 67.16 | [55.39; 77.96] |
| McGinity 2017 | 18 | 33 | 54.55 | [37.25; 71.32] |
| Bus 2018 | 54 | 84 | 64.29 | [53.69; 74.23] |
| Akbik 2019 | 27 | 62 | 43.55 | [31.38; 56.11] |
| Sufaro 2019 | 12 | 28 | 42.86 | [24.94; 61.73] |
| Trevisi 2020 | 64 | 147 | 43.54 | [35.60; 51.64] |
| **Total** | **277** | **555** | | |
| **Random effects model** | | | **49.30** | **[42.01; 56.61]** |

Heterogeneity: $I^2 = 63\%$, $\tau^2 = 0.0079$, $\chi^2_9 = 24.20$ ($p < 0.01$)

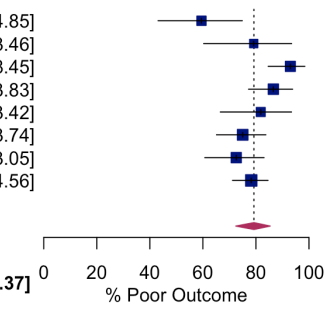



**D**

## Poor outcome at Long Term Follow Up

| Study | Poor Outcome | Total | % Poor Outcome | 95% C.I. |
|---|---|---|---|---|
| Taussky 2012 | 22 | 37 | 59.46 | [43.11; 74.85] |
| Merzo 2016 | 19 | 24 | 79.17 | [60.30; 93.46] |
| Hamed 2016 | 53 | 57 | 92.98 | [84.62; 98.45] |
| Benedetto 2017 | 58 | 67 | 86.57 | [77.22; 93.83] |
| McGinity 2017 | 27 | 33 | 81.82 | [66.57; 93.42] |
| Bus 2018 | 63 | 84 | 75.00 | [65.13; 83.74] |
| Akbik 2019 | 45 | 62 | 72.58 | [60.72; 83.05] |
| Trevisi 2020 | 115 | 147 | 78.23 | [71.17; 84.56] |
| **Total** | **402** | **511** | | |
| **Random effects model** | | | **79.25** | **[72.42; 85.37]** |

Heterogeneity: $I^2 = 66\%$, $\tau^2 = 0.0079$, $\chi^2_7 = 20.65$ ($p < 0.01$)




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**Supplementary Data 1.** Full search strategy for multi-database literature search.

**Search Strategy:**

All three searches used the same search terms and truncations. However, the three databases used a searched through a different method.

PubMed – Title/abstract
EMBASE – All fields
Cochrane Reviews – Title/abstract/keyword

**Search terms used in all three databases**

"subdural haematoma" OR "subdural hematoma" OR "subdural haemorrhage", "subdural hematoma"

**AND**

Outcome* OR Compar* OR Morbid* OR Mortality* OR Complication* OR reoccur* recur*

**AND**

Surg* OR operati* OR "non surgical" OR "non-surgical" OR "nonsurgical" OR "burrhole" OR "crani*"

**AND**

Old* OR frail* OR geri*OR elder*

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**Supplementary Data 2.** Subgroup analysis of studies published before and after 2000.

Studies published before 2000 were hypothesised to report poorer outcomes than studies published after 2000 given advances in healthcare *a priori*. Steps taken to test this are detailed below using mortality at discharge as the outcome measure.

| Study No. | Study, Year | Deaths at Discharge | Total |
|---|---|---|---|
| 1 | Wilberger 1991 [1] | 23 | 28 |
| 2 | Cagetti 1992 [2] | 23 | 26 |
| 3 | Kotwica 1992 [3] | 23 | 27 |
| 4 | Petridis 2009 [4] | 64 | 119 |
| 5 | Taussky 2012 [5] | 13 | 37 |
| 6 | Benedetto 2017 [6] | 37 | 67 |
| 7 | Won 2017 [7] | 13 | 56 |
| 8 | Monsivais 2018 [8] | 47 | 112 |
| 9 | Bus 2018 [9] | 44 | 84 |
| 10 | Akbik 2019 [10] | 24 | 62 |
| 11 | Trevisi 2020 [11] | 51 | 147 |
| 12 | Younsi 2020 [12] | 9 | 27 |

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***Identifying influential/ outlying studies***

R code output demonstrating studies in descending order with respect to their residual estimates. Studies with z-value > 1.5 were considered as potential outliers.

| | resid | se | z |
|---|---|---|---|
| 2 | 0.4342 | 0.2127 | 2.0413 |
| 3 | 0.3859 | 0.2199 | 1.7548 |
| 1 | 0.3444 | 0.2254 | 1.5277 |
| 7 | -0.3219 | 0.2179 | -1.4774 |
| 11 | -0.1899 | 0.2284 | -0.8318 |
| 12 | -0.1975 | 0.2440 | -0.8096 |
| 5 | -0.1800 | 0.2402 | -0.7496 |
| 10 | -0.1429 | 0.2372 | -0.6026 |
| 8 | -0.1084 | 0.2357 | -0.4597 |
| 6 | 0.0357 | 0.2407 | 0.1484 |
| 4 | 0.0204 | 0.2380 | 0.0857 |
| 9 | 0.0050 | 0.2397 | 0.0210 |

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***Leave-one-out tests to ascertain whether outliers are influential***

| | estimate | zval | pval | ci.lb | ci.ub | Q | Qp | tau2 | I2 | H2 |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 0.490632 | 12.417732 | 0.000000 | 0.367038 | 0.614786 | 73.368265 | 0.000000 | 0.038132 | 91.171885 | 11.327446 |
| 2 | 0.483338 | 13.222985 | 0.000000 | 0.368397 | 0.599144 | 67.141710 | 0.000000 | 0.032437 | 89.808784 | 9.812371 |
| 3 | 0.487247 | 12.750372 | 0.000000 | 0.367424 | 0.607788 | 70.593036 | 0.000000 | 0.035606 | 90.617619 | 10.658275 |
| 4 | 0.517500 | 11.346678 | 0.000000 | 0.377123 | 0.656538 | 86.566590 | 0.000000 | 0.049536 | 92.163460 | 12.760734 |
| 5 | 0.533819 | 11.975985 | 0.000000 | 0.397708 | 0.667514 | 86.933142 | 0.000000 | 0.046334 | 92.524143 | 13.376393 |
| 6 | 0.516182 | 11.374204 | 0.000000 | 0.376397 | 0.654740 | 87.156905 | 0.000000 | 0.049256 | 92.633846 | 13.575606 |
| 7 | 0.545648 | 13.087135 | 0.000000 | 0.419060 | 0.669424 | 74.186796 | 0.000000 | 0.039037 | 91.016112 | 11.131038 |
| 8 | 0.528963 | 11.623067 | 0.000000 | 0.389688 | 0.666066 | 87.768239 | 0.000000 | 0.048445 | 92.068686 | 12.608251 |
| 9 | 0.518889 | 11.377863 | 0.000000 | 0.378617 | 0.657719 | 88.032457 | 0.000000 | 0.049508 | 92.499065 | 13.331671 |
| 10 | 0.531523 | 11.784654 | 0.000000 | 0.393656 | 0.667077 | 87.254073 | 0.000000 | 0.047480 | 92.431292 | 13.212294 |
| 11 | 0.536202 | 12.018474 | 0.000000 | 0.400142 | 0.669679 | 77.809426 | 0.000000 | 0.045792 | 91.306256 | 11.502524 |
| 12 | 0.534470 | 12.069039 | 0.000000 | 0.399278 | 0.667234 | 87.048828 | 0.000000 | 0.045823 | 92.553793 | 13.429656 |

| | rstudent | dffits | cook.d | cov.r | tau2.del | QE.del | hat | weight | dfbs | inf |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1.5277 | 0.4540 | 0.1835 | 0.9554 | 0.0381 | 73.3683 | 0.0776 | 7.7642 | 0.4562 | |
| 2 | 2.0413 | 0.6096 | 0.2917 | 0.8271 | 0.0324 | 67.1417 | 0.0767 | 7.6678 | 0.6174 | |
| 3 | 1.7548 | 0.5229 | 0.2306 | 0.8984 | 0.0356 | 70.5930 | 0.0772 | 7.7175 | 0.5271 | |
| 4 | 0.0857 | 0.0161 | 0.0003 | 1.2231 | 0.0495 | 86.5666 | 0.0889 | 8.8917 | 0.0161 | |
| 5 | -0.7496 | -0.2277 | 0.0543 | 1.1421 | 0.0463 | 86.9331 | 0.0809 | 8.0874 | -0.2275 | |
| 6 | 0.1484 | 0.0353 | 0.0014 | 1.2133 | 0.0493 | 87.1569 | 0.0859 | 8.5907 | 0.0353 | |
| 7 | -1.4774 | -0.4374 | 0.1720 | 0.9838 | 0.0390 | 74.1868 | 0.0846 | 8.4625 | -0.4371 | |
| 8 | -0.4597 | -0.1513 | 0.0252 | 1.1985 | 0.0484 | 87.7682 | 0.0887 | 8.8666 | -0.1517 | |
| 9 | 0.0210 | -0.0040 | 0.0000 | 1.2205 | 0.0495 | 88.0325 | 0.0873 | 8.7273 | -0.0040 | |
| 10 | -0.6026 | -0.1909 | 0.0392 | 1.1731 | 0.0475 | 87.2541 | 0.0854 | 8.5376 | -0.1910 | |
| 11 | -0.8318 | -0.2643 | 0.0727 | 1.1406 | 0.0458 | 77.8094 | 0.0897 | 8.9692 | -0.2646 | |
| 12 | -0.8096 | -0.2388 | 0.0591 | 1.1264 | 0.0458 | 87.0488 | 0.0772 | 7.7175 | -0.2385 | |

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***Moderator analysis (using pre/post 2000 as moderator and logit transformation for each group)***

Fixed-Effects with Moderators Model (k = 2)

I^2 (residual heterogeneity / unaccounted variability): 0.00%
H^2 (unaccounted variability / sampling variability): 1.00

Test for Residual Heterogeneity:
QE(df = 0) = 0.000, p-val = 1.000

Test of Moderators (coefficient 2):
QM(df = 1) = 48.197, p-val < .001

Model Results:

| | estimate | se | zval | pval | ci.lb | ci.ub | |
|---|---|---|---|---|---|---|---|
| intrcpt | 0.701 | 0.037 | 19.132 | <.001 | 0.629 | 0.773 | *** |
| studyyearpre2000 | 0.459 | 0.066 | 6.942 | <.001 | 0.329 | 0.589 | *** |

---
Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

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**Supplementary Data 3.** Indications for conservative management in included studies, and subsequent mortality/ outcomes.

| Author, Year | No. of Patients | *At Discharge* (N, %) | | | *At Long Term Follow Up* (N, %) | | | | Indications |
|---|---|---|---|---|---|---|---|---|---|
| | | Deaths | GOS 1 – 3 | GOS 4 – 5 | Deaths | GOS 1 – 3 | GOS 4 – 5 | Duration | |
| Taussky et al., 2012[5] | 5 | - | - | - | - | 5 | 0 | 6 | Any patients not fulfilling criteria for surgery (see **Supplementary Table 2**), or with bilaterally fixed pupils. |
| Won et al., 2017[7] | 7 | - | - | - | 5 | 7 | 0 | 3 | Not specified. Seven out of 68 patients were managed conservatively due to inoperable comorbidities. |
| Sufaro et al., 2019[13] | 26 | 2 | - | - | 9 | - | - | 12 | No specific indications for conservative management given. No significant differences between surgical and conservative groups except for ASDH thickness and focal neurological deficit. |
| Trevisi et al., 2020[11] | 66 | 14 | 21 | 45 | 21 | 27 | 39 | 6 | No specific indications for conservative management. Significantly larger proportion of conservatively managed patients in GCS 13 - 15 category (77%) compared with surgical (36%) group. |

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**Supplementary Data 4.** Indications for surgical intervention in included studies.

| Author, Year | Indication/ Prerequisites |
|---|---|
| Wilberger et al., 1991[1] | Surgical indication not specified.<br>All included patients had ASDH thickness >= 3mm and MLS >5 mm.<br>Following patients were excluded from analyses:<br>● Timing of surgery could not be determined<br>● Open head injuries, prolonged hypotension/ hypoxia, severe life-threatening extra-cranial injuries |
| Massaro et al., 1996[14] | Surgical indications- MLS > 5 mm<br>Following patients were excluded- open head injury, prolonged hypotension, hypoxia or severe life threatening injuries |
| Koc et al., 1997[15] | Surgical indications- ASDH thickness > 10mm, MLS >5 mm |
| Taussky et al., 2012[5] | Following patients considered suitable for surgical intervention:<br>● Pre-morbid functional status- KPS of at least 80, usually independent<br>● Co-morbidities- no evidence of dementia, no comorbidities limiting survival to less than 12 months<br>● Surgical indications- MLS =>5mm; GCS =<13 |
| Merzo et al., 2016[16] | Not specified for ASDH patients, though general indications for surgical intervention in TBI patients given |
| Raj et al., 2016[17] | Brain trauma foundation guidelines |
| Benedetto et al., 2017[6] | Following patients considered suitable for surgical intervention:<br>● ASDH thickness > 10 mm or MLS >5 mm with patient of any GCS<br>● GCS drop of 2 points or more, with less severe ASDH thickness/ MLS<br>Following patients were excluded from analysis:<br>● Bilateral fixed pupils<br>● Concomitant EDH or significant cerebral contusions<br>● Major thoracic or abdominal trauma |
| Monsivais et al., 2018[8] | Following patients considered unsuitable for surgical intervention:<br>● Neurologically poor- GCS 3 - 5 with or without pupillary involvement with evidence of impending herniation on CT imaging<br>● Haemodynamic instability, unsuitable for ventilation, or severe cardiac/ pulmonary compromise |

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| | |
|---|---|
| | • Advanced malignancy with metastatic disease |
| Sufaro et al., 2019[13] | Following patients considered suitable for inclusion:<br>• GCS 13 - 15 and at least one of the following- ASDH thickness > 10mm, MLS > 5 mm, GCS drop of 2 points or more from time of injury<br>• Surgical intervention performed predominantly in patients with evolving hemiparesis<br>Patients with significant major injuries were excluded. |
| Trevisi et al., 2020[11] | Following patients considered suitable for surgical intervention:<br>• ASDH thickness > 10mm, MLS >5 mm (unless other clinical features/ demographics/ baseline functional status were not in favour of surgical intervention- objective parameters not specified) |
| Cagetti et al., 1992[2] | Not specified |
| Kotwica & Jakubowski, 1992[3] | Not specified |
| Akbik et al., 2009[10] | Not specified |
| Petridis et al., 2009[4] | Not specified |
| Hamed et al., 2016[18] | Not specified |
| Won et al., 2017[7] | Not specified |
| Bus et al., 2019[9] | Not specified |
| McGinity et al., 2017[19] | Not specified |
| Younsi et al., 2020[12] | Not specified |

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**Supplementary Data 5.** Association between pre-operative neurological status and mortality/ poor outcome following surgical evacuation of ASDH in patients aged 60 years and above.

| Study | GCS | Outcome | Statistical Test |
|---|---|---|---|
| Jamjoom, 1992[20] | Dichotomised: >=5 vs <5 | Dichotomised GOS | Chi-squared test; NS |
| Petridis et al., 2009[4] | Categorised: 13-15, 9-12, 3-8 | Mortality | Chi-squared test; p < 0.001 |
| Raj et al., 2016[17] | Categorised: 13-15, 9-12, 3-8 | Mortality | Chi-squared test; p < 0.001 |
| Benedetto et al., 2017[6] | GCS | GOS (30 days) | Univariate linear regression; RC 0.18, p<0.0001 |
| | | | Multivariate linear regression; RC 0.17, p<0.0001 |
| | | GOS (6 months) | Univariate linear regression; RC 0.20, p<0.0001 |
| | | | Multivariate linear regression; RC 0.20, p<0.0001 |
| Won et al., 2017[7] | Dichotomised: >6 vs <=6 | Mortality | Univariate logistic regression; OR 4.0, p = 0.04 |
| | | Dichotomised GOS (discharge) | Univariate logistic regression; OR 4.2, NS |
| | | Dichotomised GOS (LTFU) | Univariate logistic regression; OR 3.7, NS |
| Monsivais et al., 2018[8] | Dichotomised: >=9 vs <9 | Mortality | Chi-squared test; p = 0.01<br>Multivariate logistic regression; OR 3.0, p = 0.02 |
| Akbik et al., 2019[10] | GCS (mean) | Mortality | Kruskal Wallis test; p = 0.014 |
| | | Categorised GOS | Kruskal Wallis test; p = 0.016 |

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| | | | |
|---|---|---|---|
| Bus et al., 2019[9] | Categorised: 3-8, 9-15 | Dichotomised GOS | Chi-squared test; NS |
| Trevisi et al., 2020[11] | GCS | Dichotomised GOS | Univariate logistic regression; $p < 0.001$ |
| | | | Multivariate logistic regression; OR 0.87, $p = 0.04$ |
| | Dichotomised: >8 vs <=8 | Dichotomised GOS | Chi-squared test; $p < 0.001$ |

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**Supplementary Data 6.** Association between pre-operative pupil reactivity to light and mortality/ poor outcome following surgical evacuation of ASDH in patients aged 60 years and above.

| Study | Pupils | Outcome | Statistical Test |
|---|---|---|---|
| Jamjoom, 1992[20] | Non-reactive pupil(s) | Dichotomised GOS | Chi-squared test; p = 0.025 |
| Petridis et al., 2009[4] | Bilaterally reactive, unilaterally reactive, bilaterally unreactive | Mortality<br>GOS (1 vs others) | Chi-squared test; p<0.001 |
| Raj et al., 2016[17] | Normal, abnormal | Mortality | Chi-squared test; NS |
| Akbik et al., 2019[10] | Fixed and dilated, normal | Mortality | Fisher's exact test; p = 0.021 |
| | | Categorised GOS | Fisher's exact test; p = 0.009 |
| Bus et al., 2019[9] | Bilaterally dilated and unreactive | Dichotomised GOS | Fisher's exact test; p = 0.03 |
| Trevisi et al., 2020[11] | Fixed pupils | Dichotomised GOS | Chi-squared test; NS |
| Younsi et al., 2020[12] | Anisocoria | Dichotomised GOS | Fisher's exact test; NS |

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**Supplementary Data 7.** Summary of risk of bias assessment using ROBINS-I tool for all included studies.

| | Study | Cofounding | Selection | Classification | Deviation | Missing | Measurement | Reporting | Overall |
|---|---|---|---|---|---|---|---|---|---|
| **1** | Wilberger 1991 | Moderate | Low | Low | Low | Low | Moderate | Moderate | **Moderate** |
| **2** | Cagetti 1992 | Serious | NI | Moderate | Low | Low | Serious | Moderate | **Serious** |
| **3** | Jamjoom 1992 | NI | Low | Low | Low | Low | Moderate | Moderate | **Moderate** |
| **4** | Kotwica 1992 | Moderate | Low | Low | Low | Low | Moderate | Low | **Moderate** |
| **5** | Massaro 1996 | Moderate | Low | Low | Low | Low | Moderate | Low | **Moderate** |
| **6** | Koc 1997 | Moderate | Low | Low | Low | Low | Moderate | Low | **Moderate** |
| **7** | Hanif 2009 | Moderate | Low | Low | Low | Low | Moderate | Low | **Moderate** |
| **8** | Petridis 2009 | Moderate | Low | Low | Low | Low | Moderate | Low | **Moderate** |
| **9** | Taussky 2012 | Moderate | Low | Low | Low | Low | Moderate | Low | **Moderate** |
| **10** | Hamed 2016 | Serious | Low | Low | Moderate | Low | Moderate | Low | **Serious** |
| **11** | Merzo 2021 | Moderate | Moderate | Low | Low | Low | Moderate | Low | **Moderate** |
| **12** | Raj 2016 | Moderate | Moderate | Low | Low | Low | Moderate | Low | **Moderate** |
| **13** | Benedetto 2017 | Moderate | Low | Low | Low | Low | Moderate | Low | **Moderate** |
| **14** | McGinity 2017 | Moderate | Moderate | Low | Low | Low | Moderate | Low | **Moderate** |
| **15** | Won 2017 | Moderate | Low | Moderate | Low | Low | Moderate | Low | **Moderate** |
| **16** | Bus 2018 | Moderate | Low | Low | Low | Low | Moderate | Moderate | **Moderate** |
| **17** | Monsivai 2018 | Moderate | Low | Low | Low | Low | Moderate | Low | **Moderate** |
| **18** | Akbik 2019 | Moderate | Low | Low | Low | Low | Moderate | Low | **Moderate** |
| **19** | Sufaro 2019 | Serious | Low | Low | Moderate | Low | Moderate | Low | **Serious** |
| **20** | Trevisi 2020 | Moderate | Low | Low | Low | Low | Moderate | Low | **Moderate** |
| **21** | Younsi 2020 | Moderate | Low | Low | Low | Serious | Moderate | Low | **Serious** |

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# PRISMA 2020 Checklist

| Section and Topic | Item # | Checklist item | Location where item is reported |
|---|---|---|---|
| **TITLE** | | | |
| Title | 1 | Identify the report as a systematic review. | Page 1 |
| **ABSTRACT** | | | |
| Abstract | 2 | See the PRISMA 2020 for Abstracts checklist. | Abstracts checklist attached |
| **INTRODUCTION** | | | |
| Rationale | 3 | Describe the rationale for the review in the context of existing knowledge. | Page 3 (Introduction) |
| Objectives | 4 | Provide an explicit statement of the objective(s) or question(s) the review addresses. | Page 4 (Introduction, final paragraph) |
| **METHODS** | | | |
| Eligibility criteria | 5 | Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses. | Page 5 (Methods, Study Selection) |
| Information sources | 6 | Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted. | Page 5-6 (Methods, Literature Search) |
| Search strategy | 7 | Present the full search strategies for all databases, registers and websites, including any filters and limits used. | Pages 5-6 (Methods); Supplementary Data 1 |
| Selection process | 8 | Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process. | Pages 4-5 (Methods) |
| Data collection process | 9 | Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process. | Pages 4-5 (Methods) |
| Data items | 10a | List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect. | Page 5 (Methods) |
| | 10b | List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information. | Page 5 (Methods) |
| Study risk of bias assessment | 11 | Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process. | Page 6 (Methods) |
| Effect measures | 12 | Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results. | Page 6 (Methods) |

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# PRISMA 2020 Checklist

| Section and Topic | Item # | Checklist item | Location where item is reported |
|---|---|---|---|
| Synthesis methods | 13a | Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)). | Page 6 (Methods) |
| | 13b | Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions. | N/A |
| | 13c | Describe any methods used to tabulate or visually display results of individual studies and syntheses. | Page 6 (Methods) |
| | 13d | Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used. | Page 6 (Methods) |
| | 13e | Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression). | Page 6 (Methods); Supplementary Data 2 |
| | 13f | Describe any sensitivity analyses conducted to assess robustness of the synthesized results. | N/A |
| Reporting bias assessment | 14 | Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases). | Page 6 (Methods) |
| Certainty assessment | 15 | Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome. | N/A |
| **RESULTS** | | | |
| Study selection | 16a | Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram. | Figure 1 |
| | 16b | Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded. | Figure 1 |
| Study characteristics | 17 | Cite each included study and present its characteristics. | Table 1 |
| Risk of bias in studies | 18 | Present assessments of risk of bias for each included study. | Table 1, Supplementary Table 7 |
| Results of individual studies | 19 | For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots. | Page 7 (Results); Figure 2 |
| Results of syntheses | 20a | For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies. | Table 1 |
| | 20b | Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect. | Page 7 (Results); Figure 2 |
| | 20c | Present results of all investigations of possible causes of heterogeneity among study results. | Supplementary Data 2 |
| | 20d | Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results. | N/A |
| Reporting biases | 21 | Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed. | Table 1 |

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# PRISMA 2020 Checklist

| Section and Topic | Item # | Checklist item | Location where item is reported |
|---|---|---|---|
| Certainty of evidence | 22 | Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed. | Page 12-13 (Discussion) |
| **DISCUSSION** | | | |
| Discussion | 23a | Provide a general interpretation of the results in the context of other evidence. | Page 12-13 (Discussion) |
| | 23b | Discuss any limitations of the evidence included in the review. | Page 15-16 (Discussion) |
| | 23c | Discuss any limitations of the review processes used. | Page 15-16 (Discussion) |
| | 23d | Discuss implications of the results for practice, policy, and future research. | Page 17 |
| **OTHER INFORMATION** | | | |
| Registration and protocol | 24a | Provide registration information for the review, including register name and registration number, or state that the review was not registered. | Page 2 |
| | 24b | Indicate where the review protocol can be accessed, or state that a protocol was not prepared. | Page 2 |
| | 24c | Describe and explain any amendments to information provided at registration or in the protocol. | N/A |
| Support | 25 | Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review. | Page 17 |
| Competing interests | 26 | Declare any competing interests of review authors. | Page 17 |
| Availability of data, code and other materials | 27 | Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review. | Page 17 |

*From:* Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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**The PRISMA for Abstracts Checklist**

| TITLE | CHECKLIST ITEM | REPORTED ON PAGE # |
|---|---|---|
| 1. Title: | Identify the report as a systematic review, meta-analysis, or both. | 1 |
| **BACKGROUND** | | |
| 2. Objectives: | The research question including components such as participants, interventions, comparators, and outcomes. | 4 |
| **METHODS** | | |
| 3. Eligibility criteria: | Study and report characteristics used as criteria for inclusion. | 5 |
| 4. Information sources: | Key databases searched and search dates. | 4 |
| 5. Risk of bias: | Methods of assessing risk of bias. | 6 |
| **RESULTS** | | |
| 6. Included studies: | Number and type of included studies and participants and relevant characteristics of studies. | 7 |
| 7. Synthesis of results: | Results for main outcomes (benefits and harms), preferably indicating the number of studies and participants for each. If meta-analysis was done, include summary measures and confidence intervals. | 7 |
| 8. Description of the effect: | Direction of the effect (i.e. which group is favoured) and size of the effect in terms meaningful to clinicians and patients. | 7 |
| **DISCUSSION** | | |
| 9. Strengths and Limitations of evidence: | Brief summary of strengths and limitations of evidence (e.g. inconsistency, imprecision, indirectness, or risk of bias, other supporting or conflicting evidence) | 12 |
| 10. Interpretation: | General interpretation of the results and important implications | 12-13 |
| **OTHER** | | |
| 11. Funding: | Primary source of funding for the review. | N/A |
| 12. Registration: | Registration number and registry name. | 2 |